

34721/34722 Linear Control Design 1

Spring Semester



Teachers: Silvia Tolu - Dimitrios Papageorgiou

Lesson 2/13

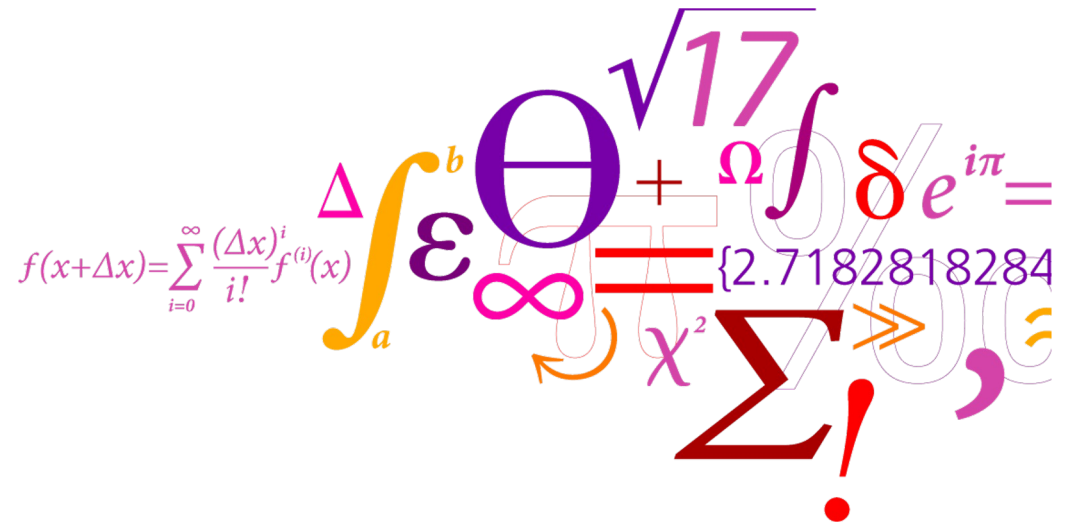
Schedule

13:00 - 15:00

- ❖ Control Intro
- ❖ Block diagram modelling
- ❖ Control process introduction
- ❖ Controller design / hand-tuning

15:00 - 17:00

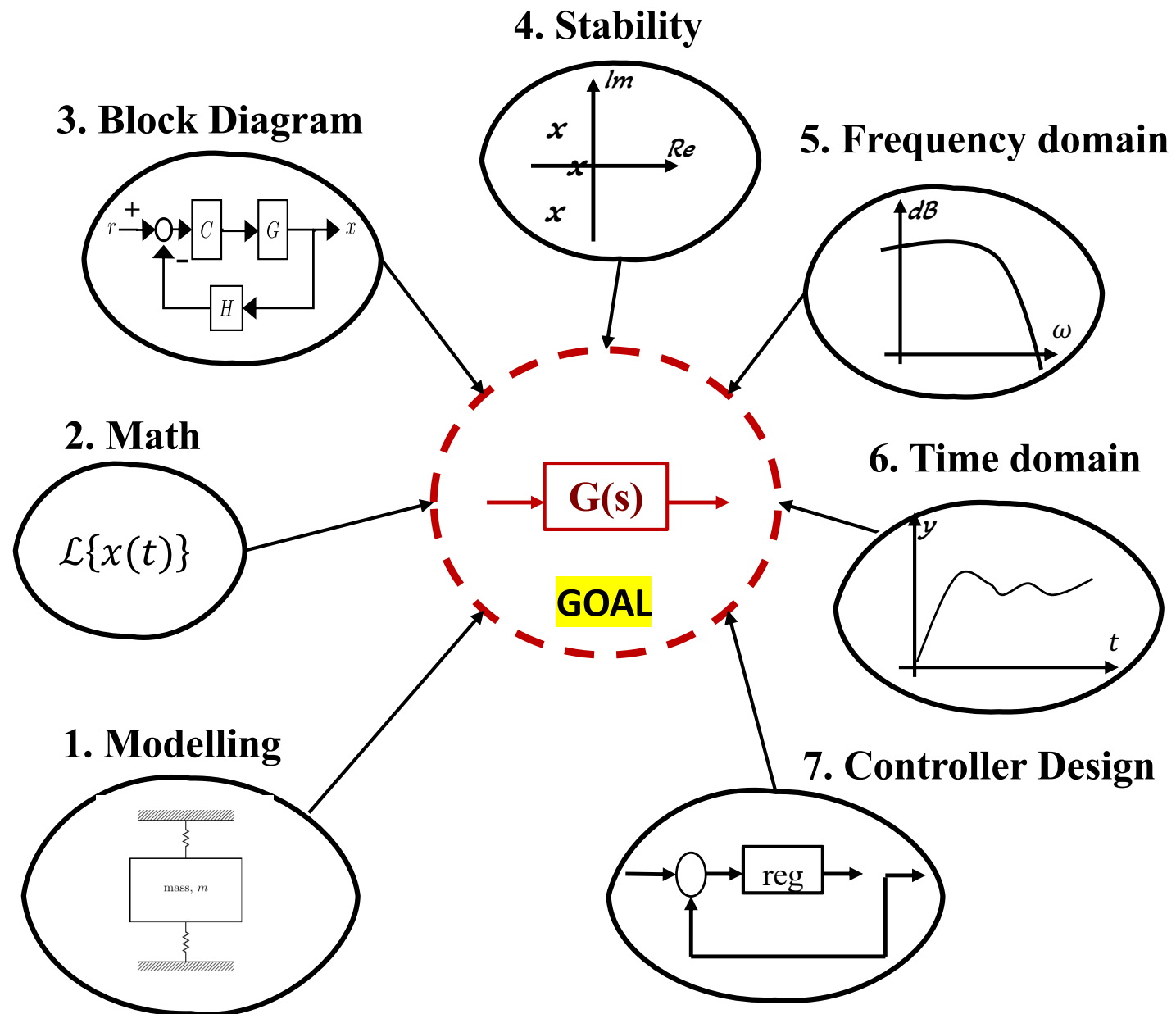
Exercise: Hand Tuning



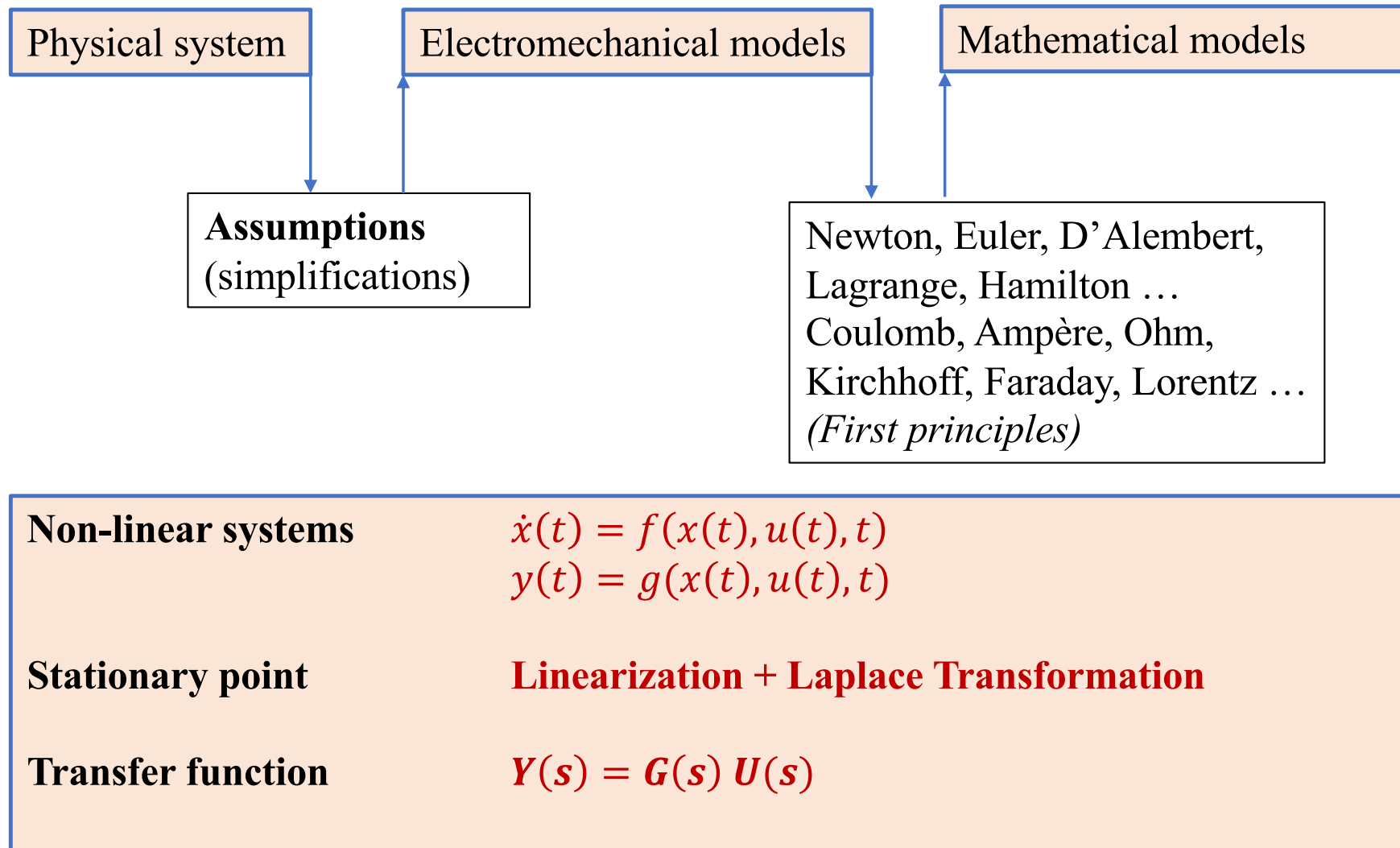
DTU ELECTRO

Department of Electrical and Photonics Engineering
Control, Robotics and Embodied AI (CREA) Group

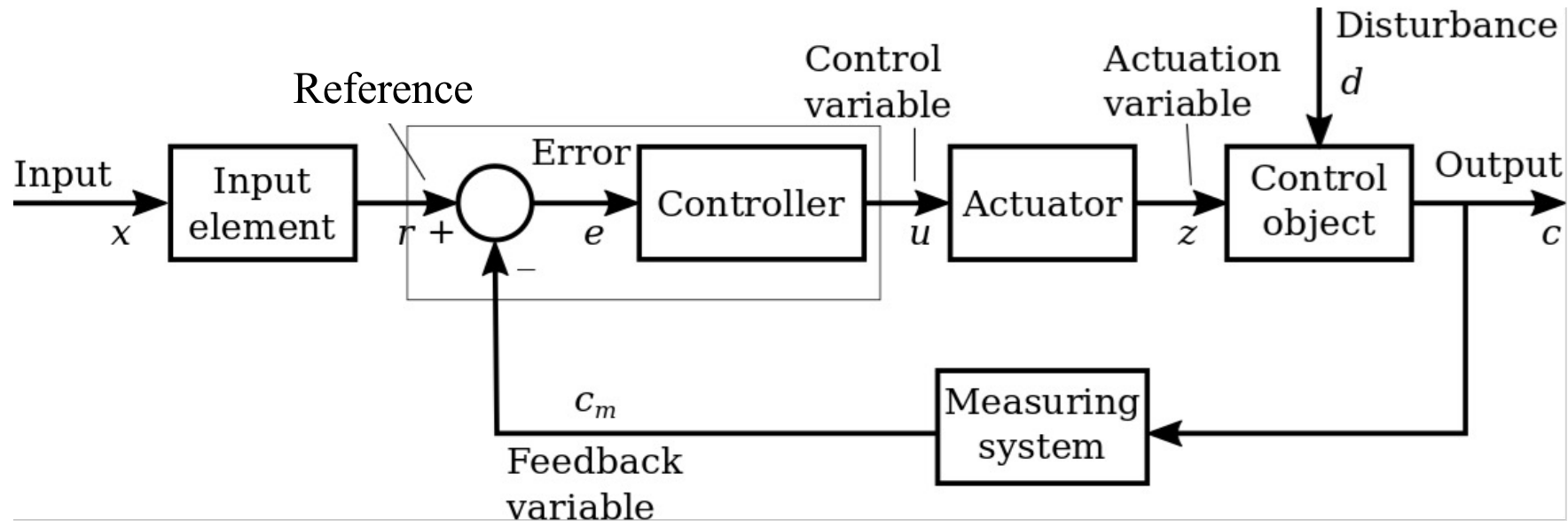
Day	Topics	Contents	Assignments
1	Introduction	Intro to feedback control, Pre-Test	MATLAB and Simulink
→ 2	Control concepts	Block diagram, control process, hand tuning	Hand tuning, Ziegler-Nichols
3	Transfer function and Laplace	Laplace and transfer function	Block diagrams and transfer functions
4	Frequency domain	Frequency domain properties	Introduction to REGBOT
5	Modelling and Linearization	White/black box modelling and linearization	REGBOT – Transfer function from data
6	Bode Plots and stability	Bode plot, stability margins, P-design	REGBOT – Bode, stability, P-Controller design



Modelling and applications of models



Typical Diagram for a Control System



Control object: **robot (balance)**

Actuator: **motor/servo**

Control variable: **voltage**

Actuation variable: **velocity/position**

Feedback: **measured variable**

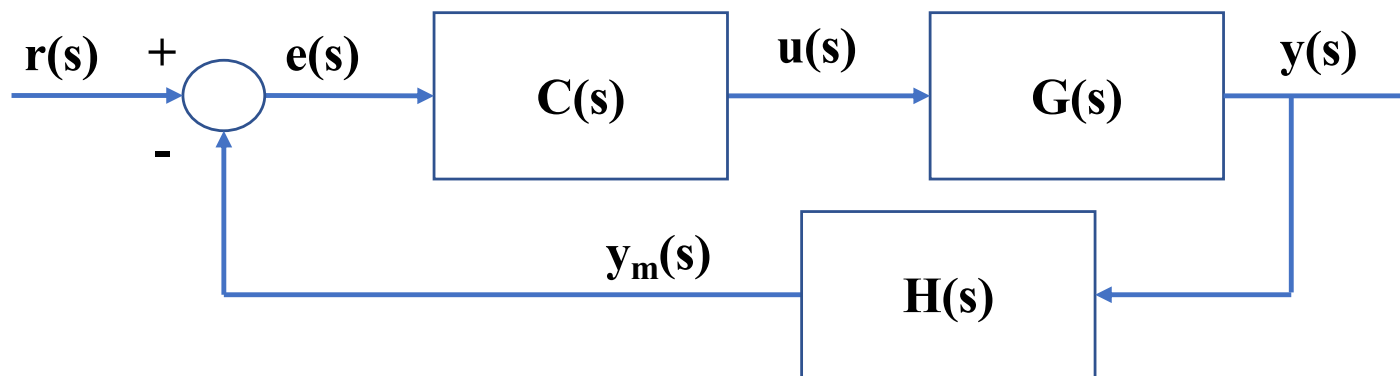
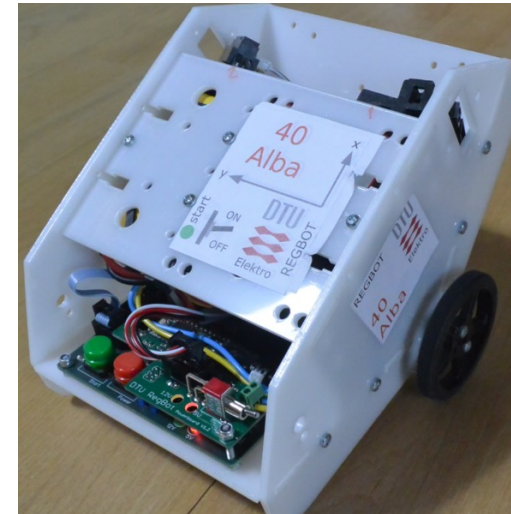


<https://rsewiki.electro.dtu.dk/index.php?title=Regbot>

GOAL

Get a mathematical representation by functions of s

- $G(s)$ - A dynamic system to be controlled
- $H(s)$ - A measure or signal (*e.g.*, the position, encoders, cameras, etc.)
- $R(s)$ - A reference value or signal for the system to follow
- $C(s)$ - the controller which provides the control signal ($\mathbf{u}$) to $\mathbf{G}$ calculated from error ($\mathbf{e}$)



Basic notions: Newton's Laws

*1st “A body continues in its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by forces impressed upon it.” (Principle of **INERTIA**)*

*2nd “The change of **motion/momentum** (ρ) of an object is proportional to the force impressed, and it is made in the direction of the straight line in which the force is impressed.*

$$\vec{\rho} = m \vec{v} \text{ (momentum)}$$
$$F = \frac{d\vec{\rho}}{dt} = m \vec{a} \text{ (force)}$$

3rd “To every action, there is always opposed an equal reaction; or the mutual actions of two bodies upon each other are always equal and directed to contrary parts.”

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- ❖ **Block diagram modelling**
- ❖ Control process introduction
- ❖ Controller design / hand-tuning

15:00 - 17:00

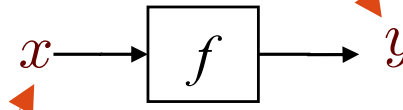
- ❖ Exercise: Hand Tuning

Linear Block Diagram

Basic elements:

Signals (x , y)

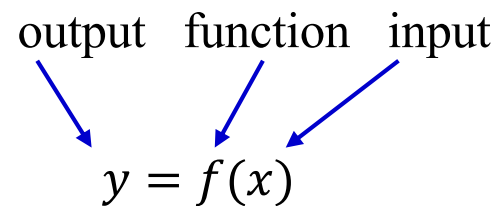
Input and output



Example:

x is input (or state)

y is output



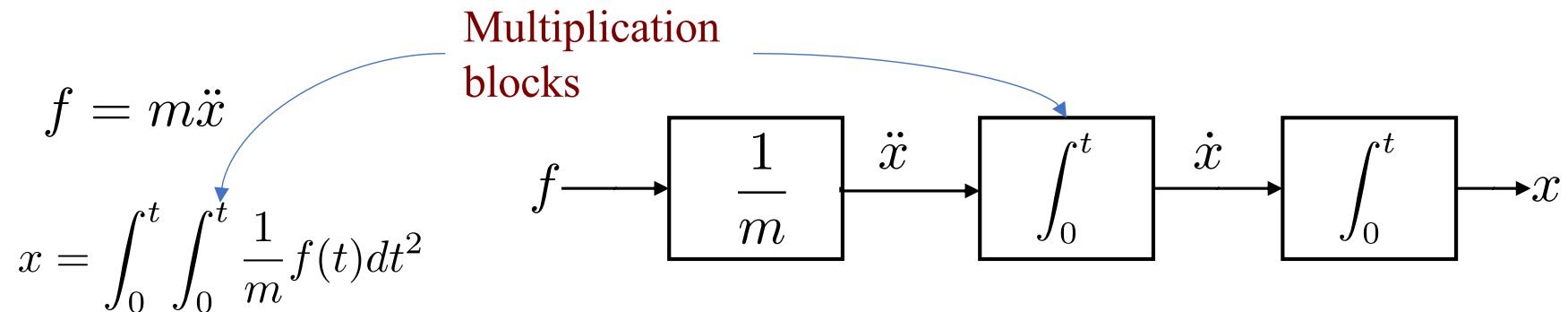
The input or state is a meaningful quantity that shows how a system evolves

The output is a quantity that we can measure and want to control

Block Diagram

- A block diagram provides a **system overview**
- Blocks describe the relationship between input and output (**Transfer Function - TF**)
- A block diagram is the link **between physics and mathematics**
- The arrows between blocks indicate **physical variables**
- Many blocks can be **easily reduced** to a few

Example: Newton's second law of motion:

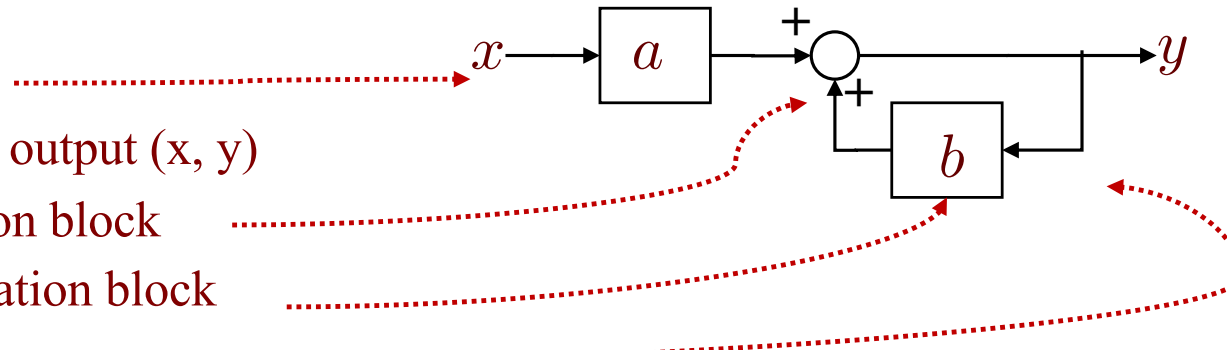


Linear Block Diagram

$$y = ax + by$$

Basic elements:

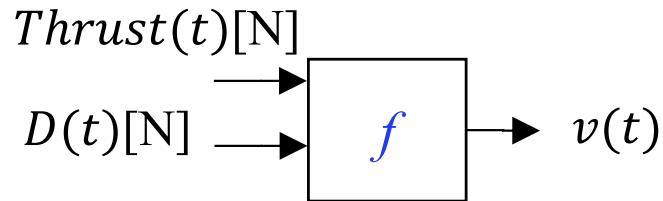
- Signals
- Input and output (x, y)
- Summation block
- Multiplication block
- Branching



Integration is allowed

It can be extended to nonlinear blocks, e.g., *limits*

Block Diagram - Example



output function input

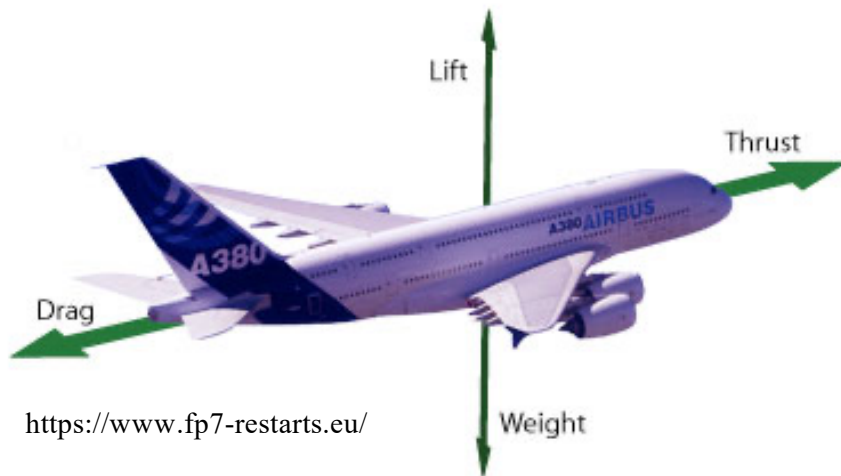
$v(t) = f(Thrust(t) - D(t))$

Blue arrows point from the labels 'output', 'function', and 'input' to $v(t)$, f , and $Thrust(t) - D(t)$ respectively.

The drag $D(t)$ is a mechanical force proportional to the speed

(the height is constant): $D(t) = k v(t)$

2nd Newton's Law: $F = m a \Rightarrow a = \frac{F}{m}$



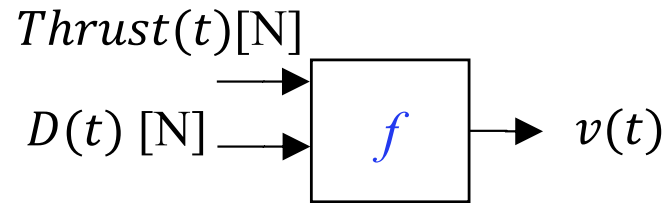
<https://www.fp7-restarts.eu/>

$$v(t) = \int_0^t \frac{1}{m} (Thrust(t) - D(t)) dt + v_0$$

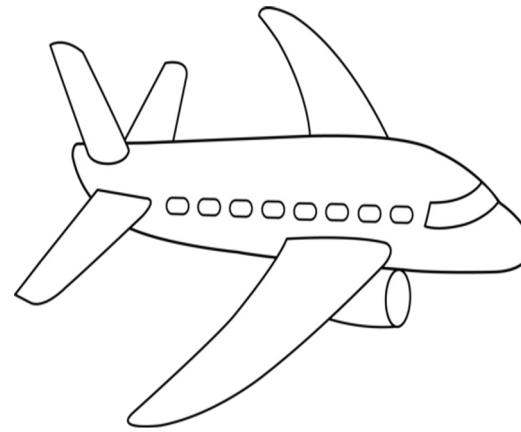
We disregard the initial value
(it starts at speed 0 at high altitude)

A curved arrow points from this text to the v_0 term in the equation above.

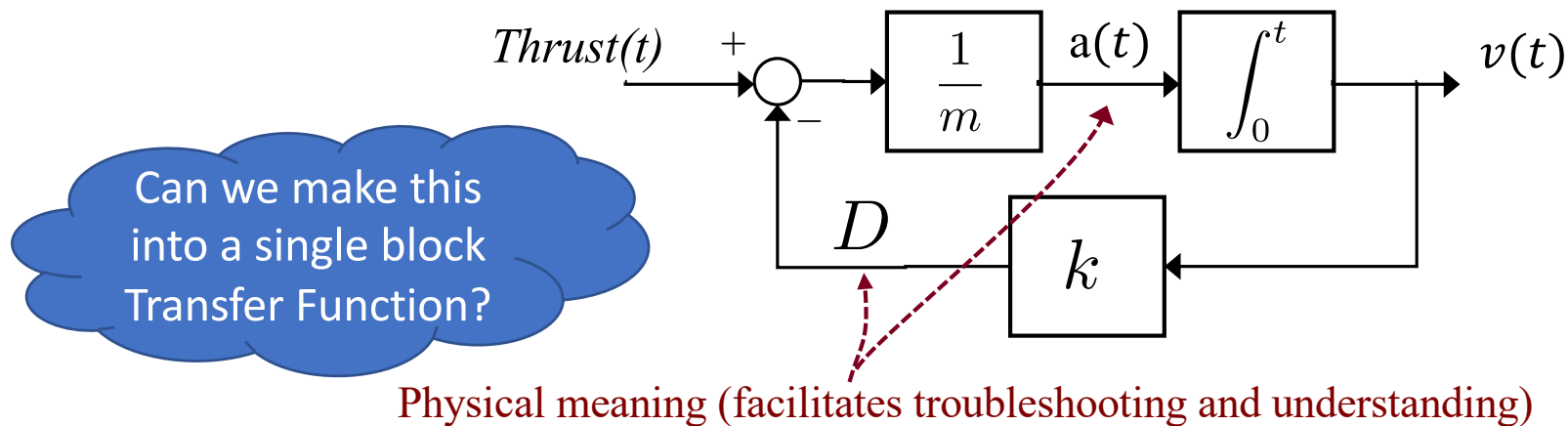
Block Diagram



$$v(t) = \int_0^t \frac{1}{m} (Thrust(t) - D(t)) dt + v_0$$

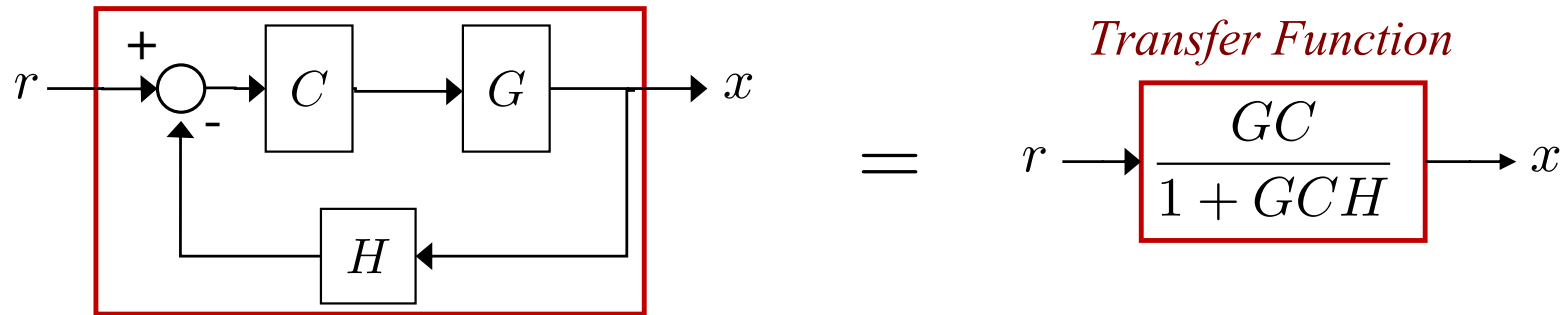


Tip: Start from the output and go backwards step by step towards the input



Block model

➔ Reduction to **Transfer Function (I)**



$$x = GC(r - Hx) \quad \leftarrow \text{Start with the output, step by step move towards input}$$

$$x = GCr - GCHx$$

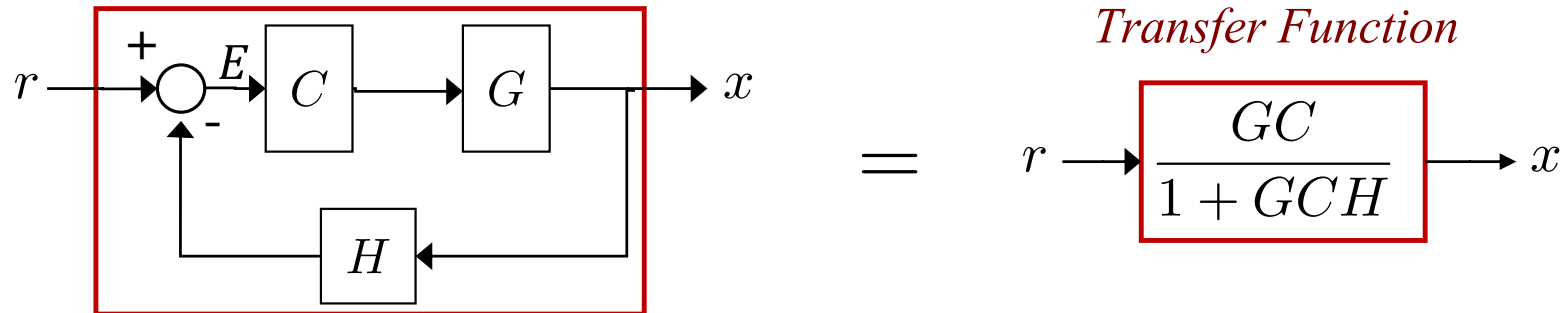
$$x + GCHx = GCr$$

$$x(1 + GCH) = GCr$$

$$\frac{x}{r} = \boxed{\frac{GC}{1 + GCH}}$$

Block model

➔ Reduction to **Transfer Function** (II)



$$x = GCE \Rightarrow E = \frac{x}{GC} \quad (1)$$

$$E = r - xH \quad (2)$$

$$\frac{x}{GC} = r - xH \quad x = (r - xH)GC \quad x(1 + GCH) = r GC$$

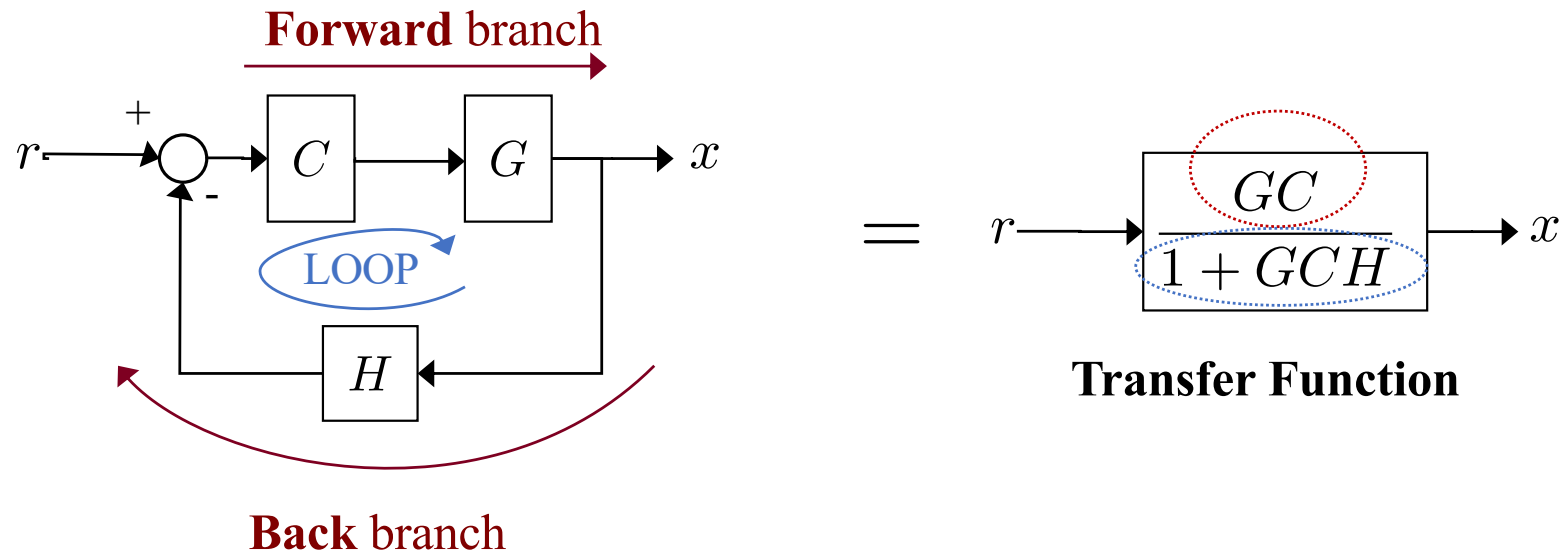
(1)

(2)

$$\frac{x}{r} = \frac{GC}{1 + GCH}$$

Block model

Reduction to transfer function - LOOP



For a loop, the following generally applies:

- The **transfer function** is a fraction
- Numerator $\rightarrow$ **forward** branch
- Denominator $\rightarrow$ 1 plus the **LOOP** (except minus)

Block Reduction

If the Laplace transformation is used, the integration is expressed as $\frac{1}{s}$

$$\boxed{\int_0^t} = \boxed{\frac{1}{s}}$$

And conversely, **s** means differentiation

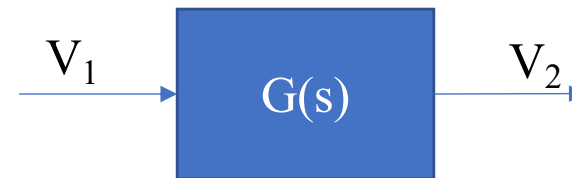
(do not confuse s with second)

We will use **s** and $\frac{1}{s}$ as a notation, and assume that we can use arithmetic rules in which **s** is a variable

(more on Laplace next lecture)

From transfer function to block diagram model (I)

$$G(s) = \frac{V_2}{V_1} = \frac{b_0}{s^2 + a_1 s + a_0}$$



Block diagram?

$$V_2(s^2 + a_1 s + a_0) = b_0 V_1$$

$$V_2 s^2 = b_0 V_1 - a_1 V_2 s - a_0 V_2$$

$$\ddot{V}_2 = b_0 V_1 - a_1 \dot{V}_2 - a_0 V_2$$

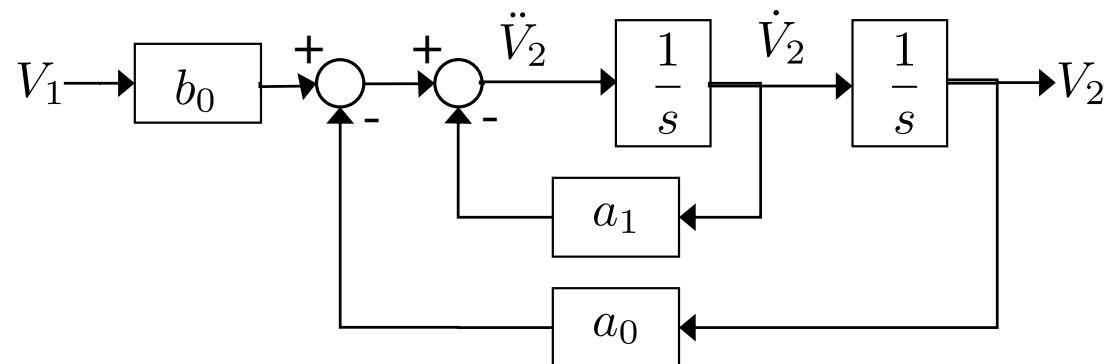
Will there be integrators?

From transfer function to block diagram model (II)

$$G(s) = \frac{V_2}{V_1} = \frac{b_0}{s^2 + a_1 s + a_0}$$

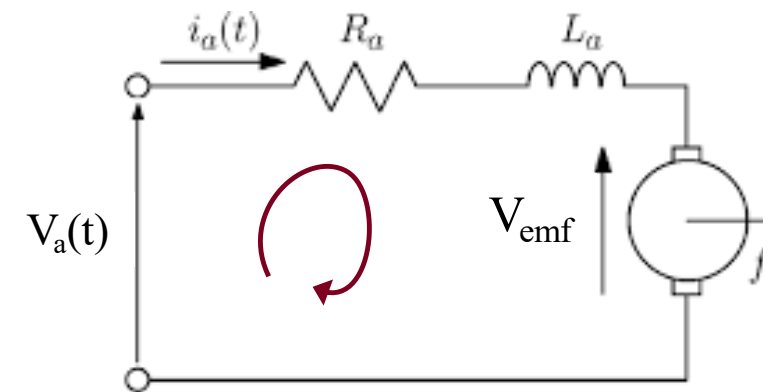
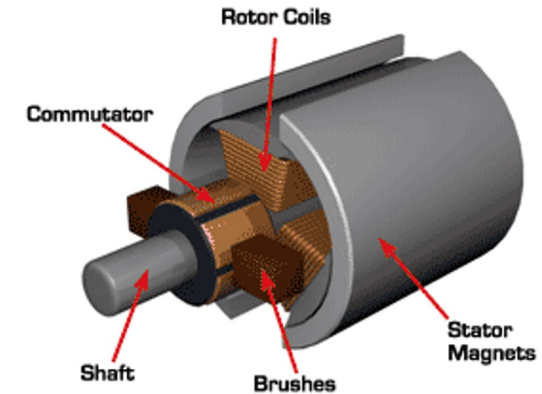

$$\ddot{V}_2 = b_0 V_1 - a_1 \dot{V}_2 - a_0 V_2$$

Leads with the highest power

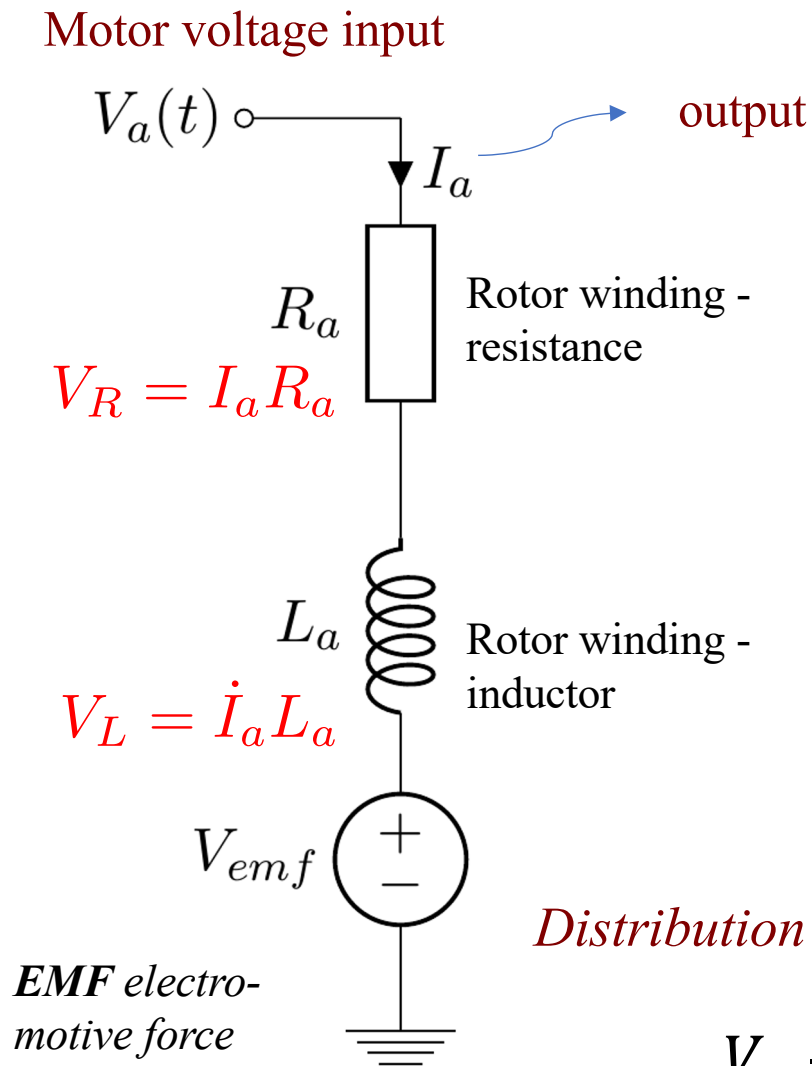


DC Motor Model – armature current I_a

Textbook Pages 99-100



Motor torque proportional to armature current $\tau_m = K_\tau I_a$



Distribution of voltage - using Kirchhoff's voltage law

$$V_a - V_R - V_L - V_{emf} = 0$$

DC Motor Model – armature current I_a

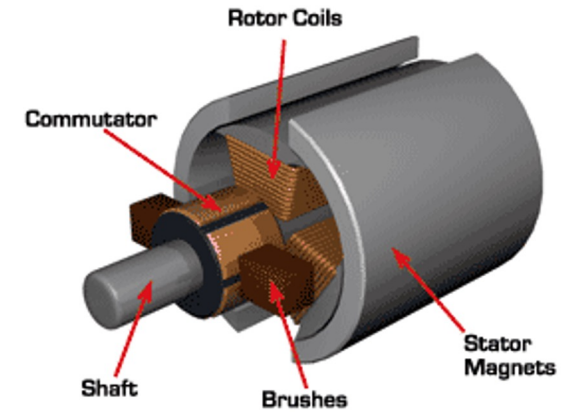
Textbook Pages 99-100

Distribution of voltage

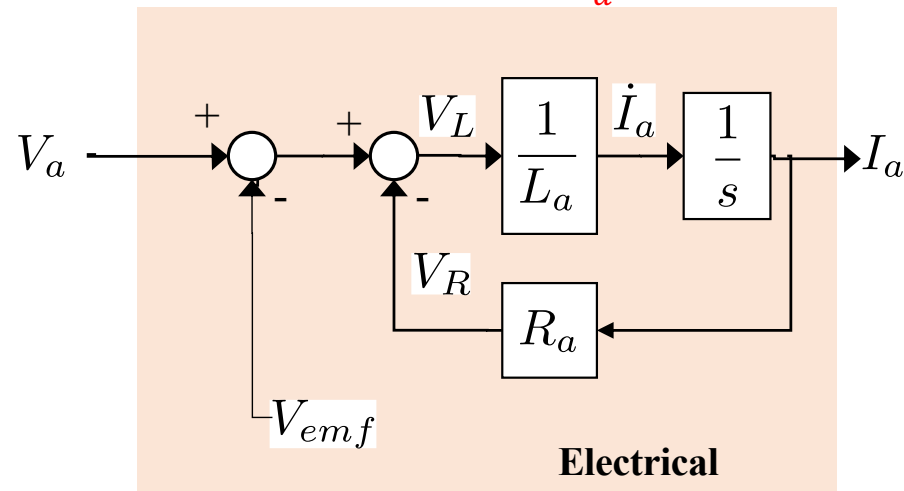
$$V_a - V_R - V_L - V_{emf} = 0$$

$$V_L = V_a - V_R - V_{emf}$$

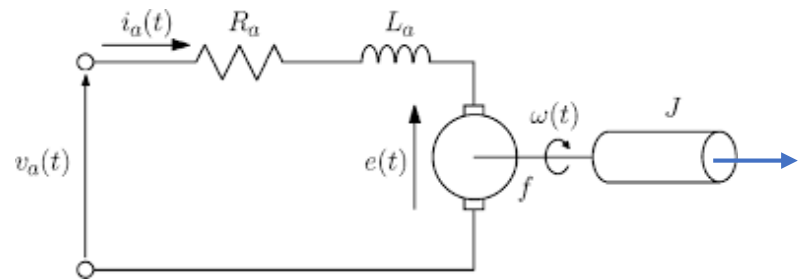
$$L_a \dot{I}_a = V_a - R_a I_a - V_{emf}$$



$$\dot{I}_a = \frac{V_L}{L_a}$$



Motor model – Electromechanical System

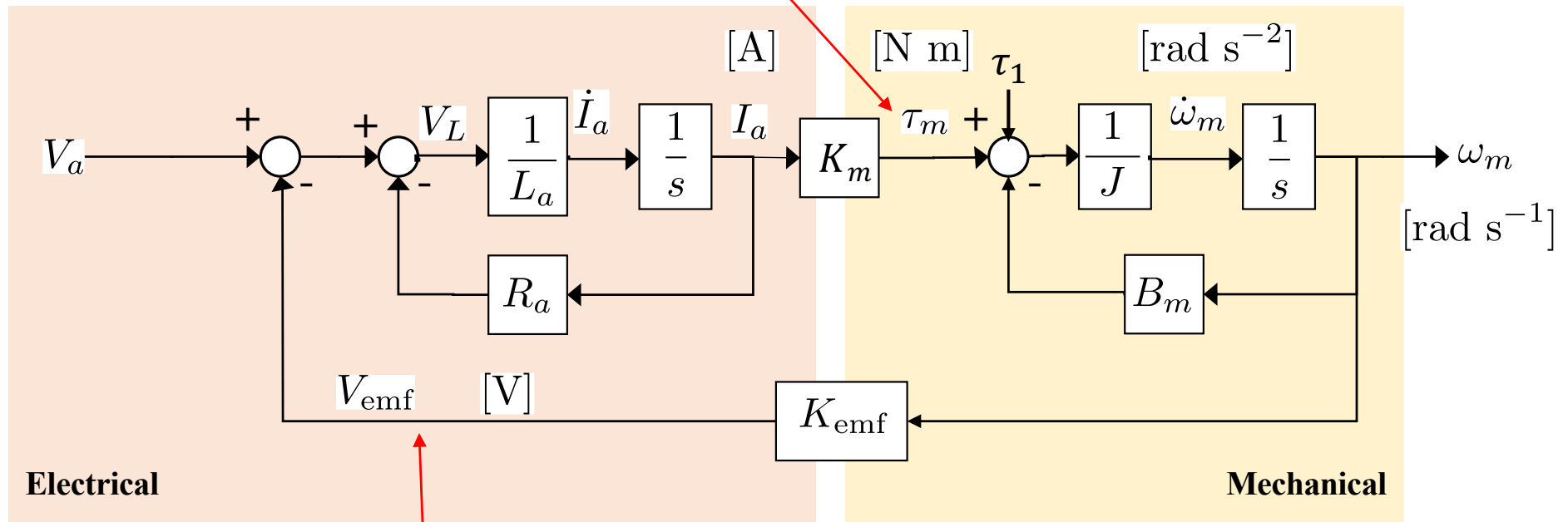


Motor torque

$$\tau_m = K_m I_a$$

Euler's 2nd law for motor inertia

$$J \dot{\omega}_m = K_m I_a - B_m \omega_m - \tau_1$$



$$V_{emf} = K_{emf} \omega_m$$

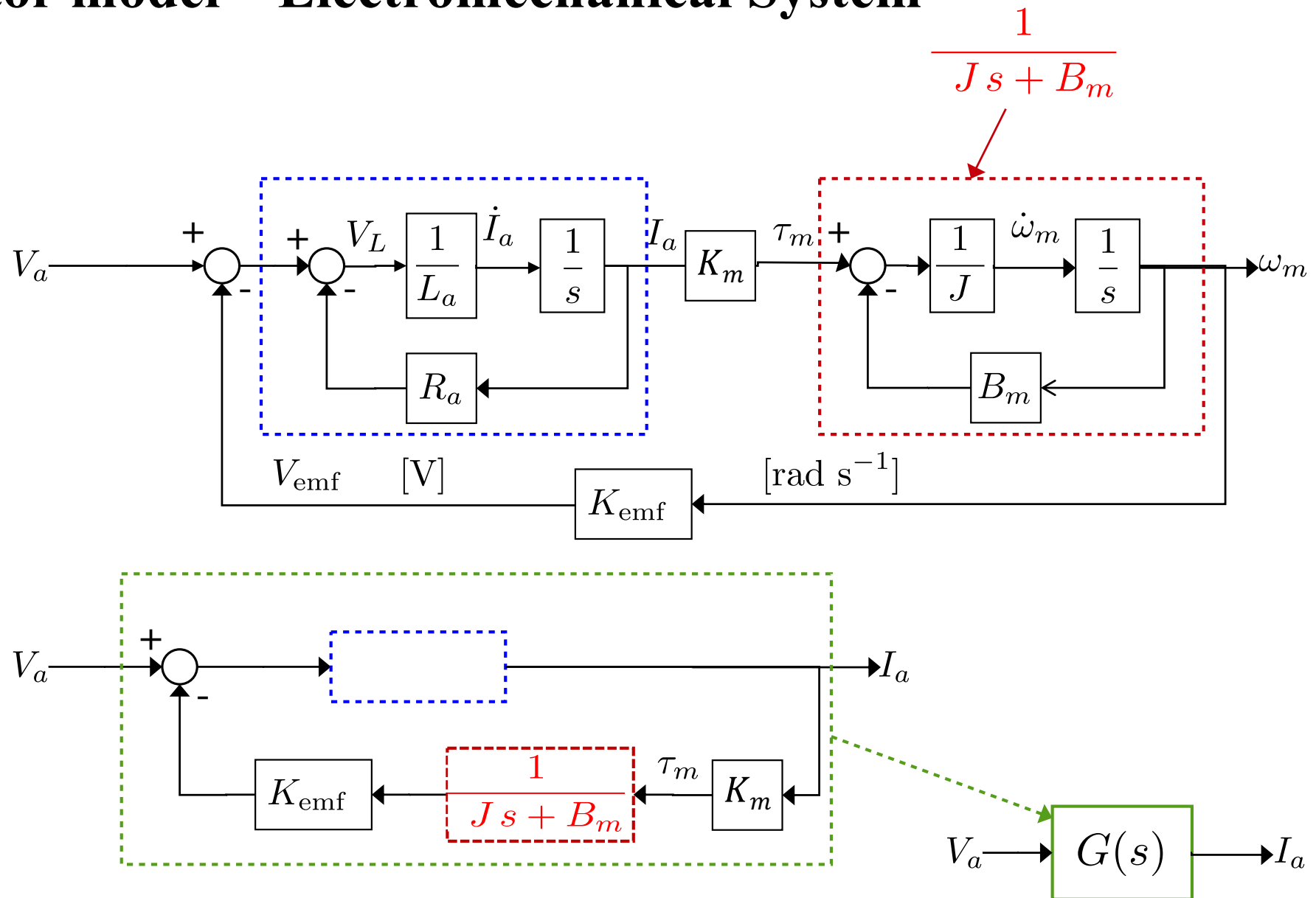
EMF electro motive force

J_m = mass moment of inertia

B_m = friction (viscose)

K_m and K_{emf} are motor constants with SI units
(Nm/A = V/(rad/s))

Motor model – Electromechanical System



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- **Control process introduction**
- Controller design - hand tuning

15:00 - 17:00

Assignment: Hand Tuning

Control Process Introduction

Heating system for premises

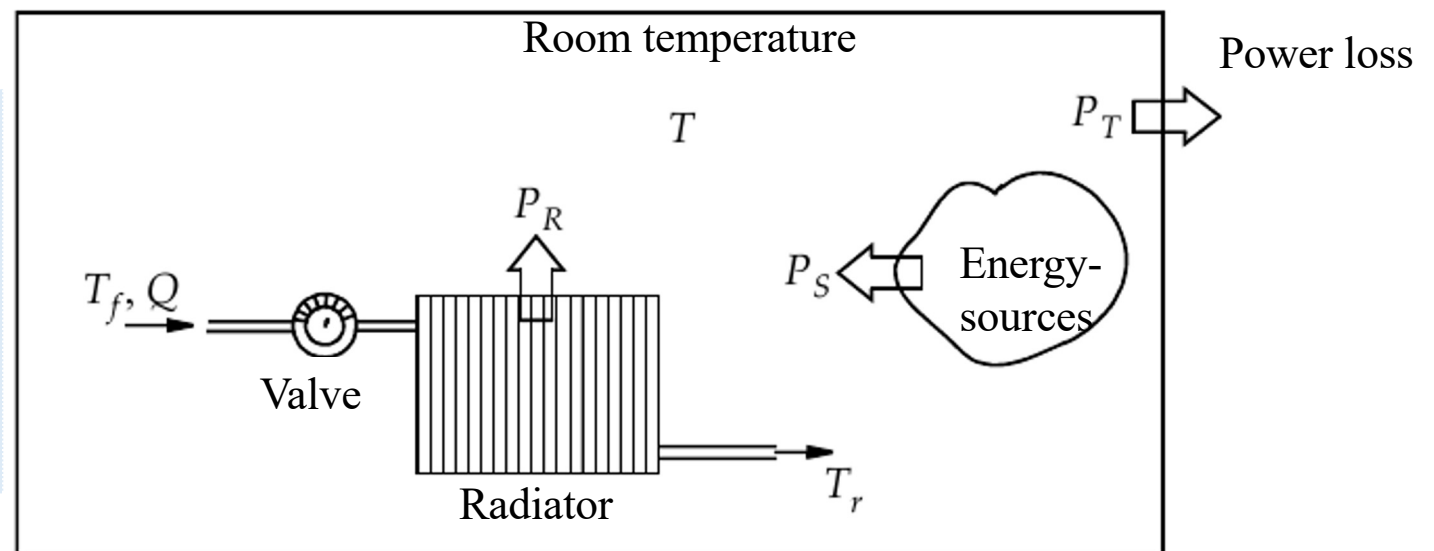
P_R power radiator
 P_S external power
(lighting, people)
 P_T drains

Input?

Output?

What do we want to control?

What are the disturbances?



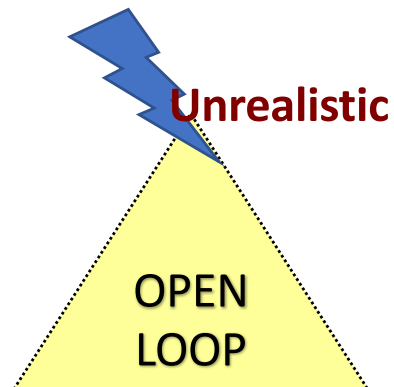
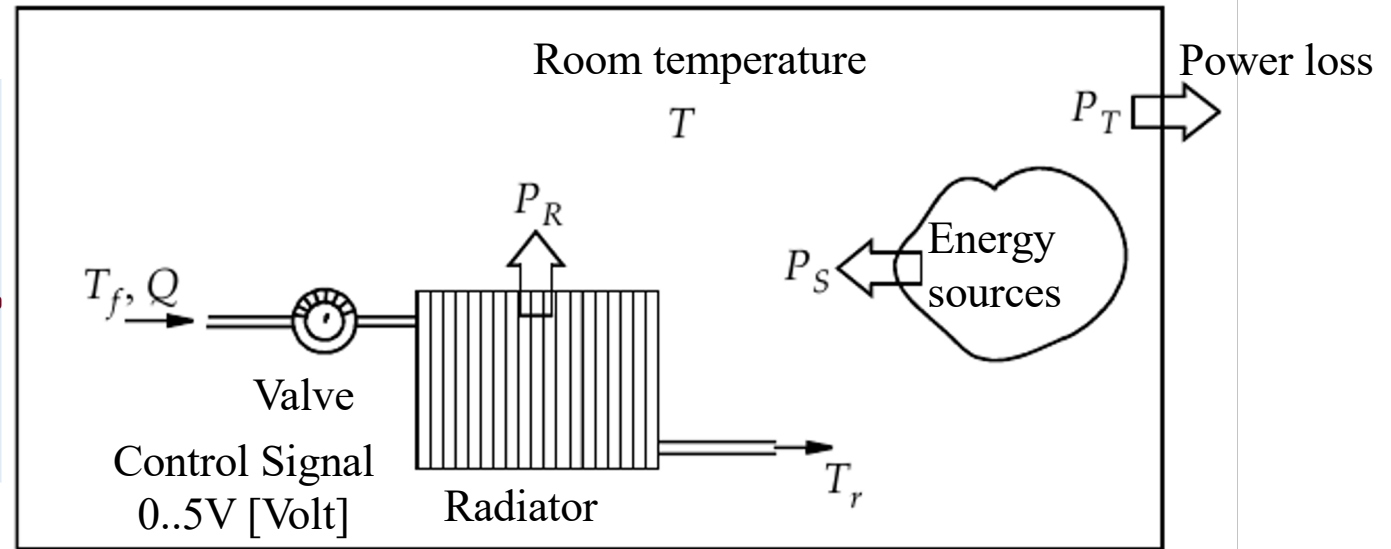
Real System - Heating system for premises

Input options?

Output?

What do we want to control?

Disturbances?



Control variable input

u

System Block Diagram

Disturbances

supplement

loss

flow
 Q

P_S

P_T

Valve

actuator

Radiator

P_R
effect

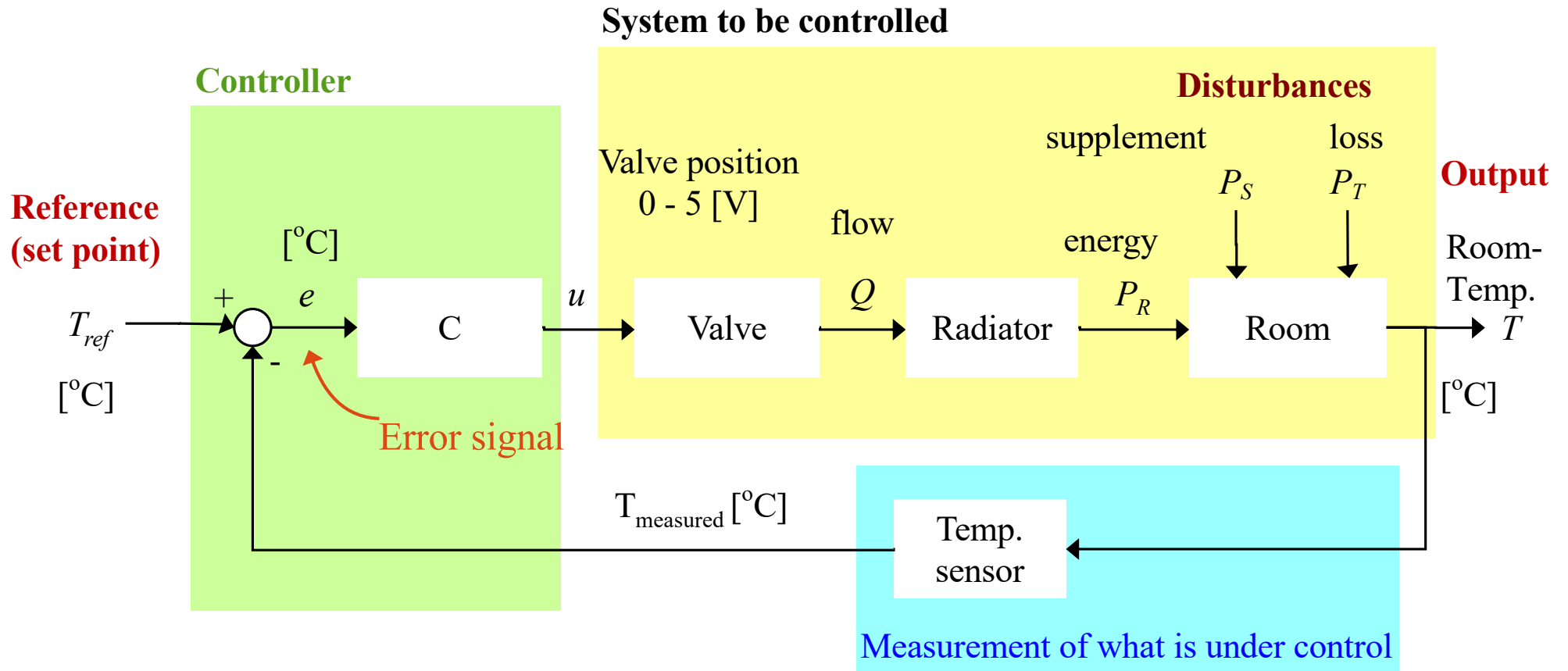
Room

Temperature

T

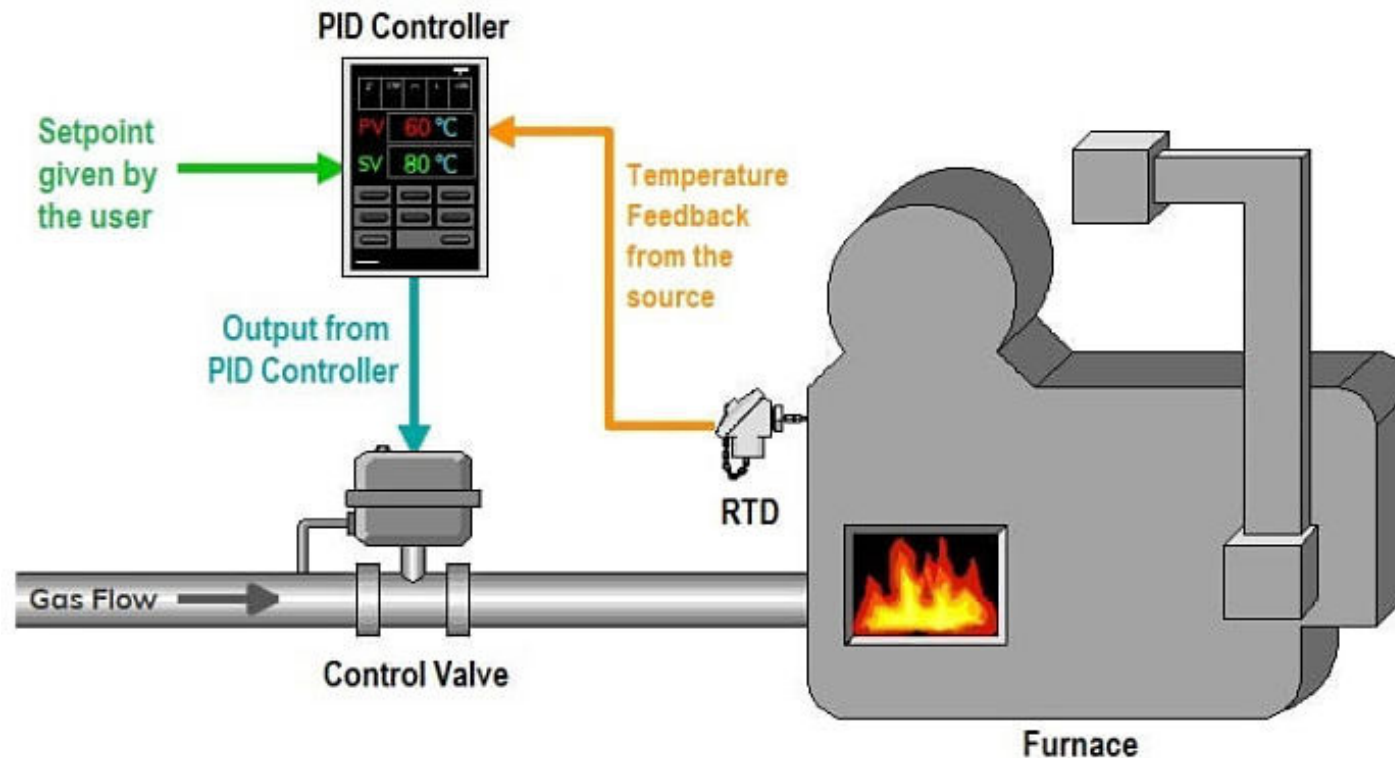
Output to control

Block Diagram – Closed-loop controller



What should the controller take care of?

BENEFITS of the **closed-loop** controller



- **Accurately tracks the reference (setpoint) input**
- **Compensates for variations and uncertainties in the system**
- **Reduces oscillations**
- **Improves system stability**
- **Rejects or significantly reduces disturbances**
- **Increases system speed**
- **Improves transient response**

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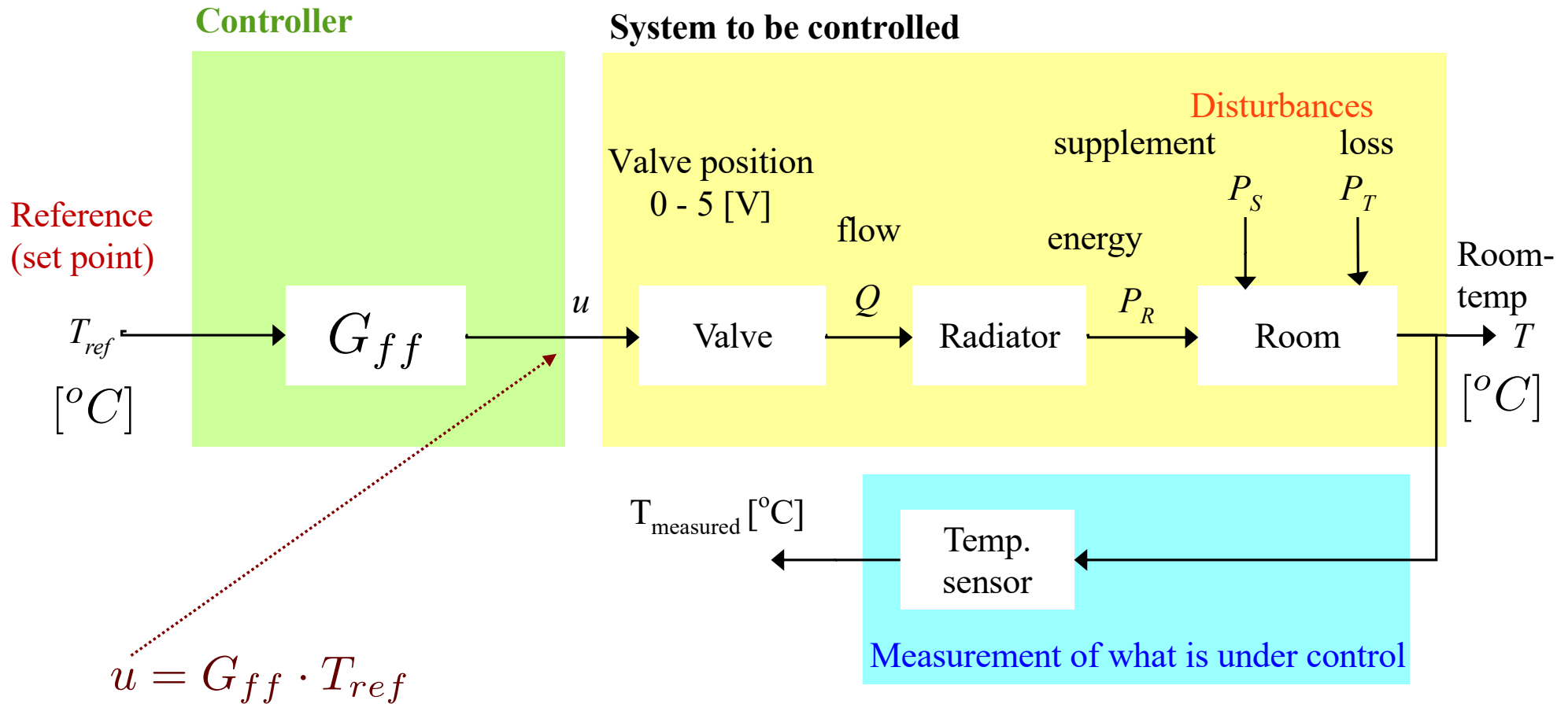
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- ❖ Control intro
- ❖ Block diagram modelling
- ❖ Control process introduction
- ❖ **Controller Design - hand tuning**

15:00 - 17:00

- ❖ Assignment: Hand Tuning

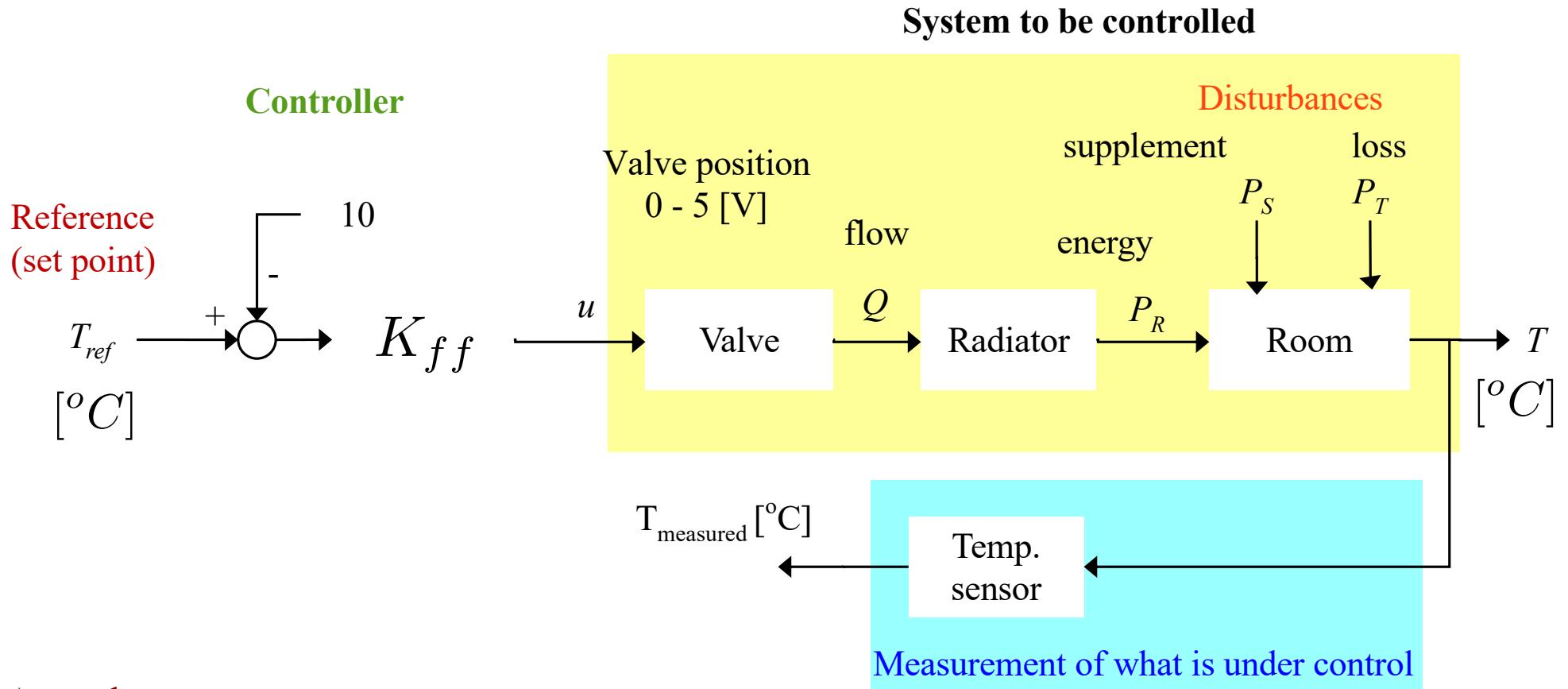
Feed-Forward (FF) Controller



$$G_{ff} = ?$$

FF Controller

It assumes a **proportional** relationship between the **input** and the **output**.
In this case, the **temperature** (T output) is assumed to be proportional to the valve control signal **u**.



A test shows

$$u = 0 \rightarrow T_{measured} = 10^{\circ}C$$

$$u = 5 \rightarrow T_{measured} = 30^{\circ}C$$

$$u = K_{ff}(T_{ref} - 10)$$

$$K_{ff} = \frac{u}{T_{ref} - 10} = \frac{5}{30 - 10} = 0.25$$

Challenges?

11.02.2026

FF Controller

Advantages

- ✓ Does not introduce instability (open-loop control)
- ✓ Constant actuator signal u for a constant reference
- ✓ Simple structure and easy to implement

Disadvantages

- ✗ Does not account for disturbances
- ✗ Cannot correct modeling errors or system variations

What about a **P-controller**?

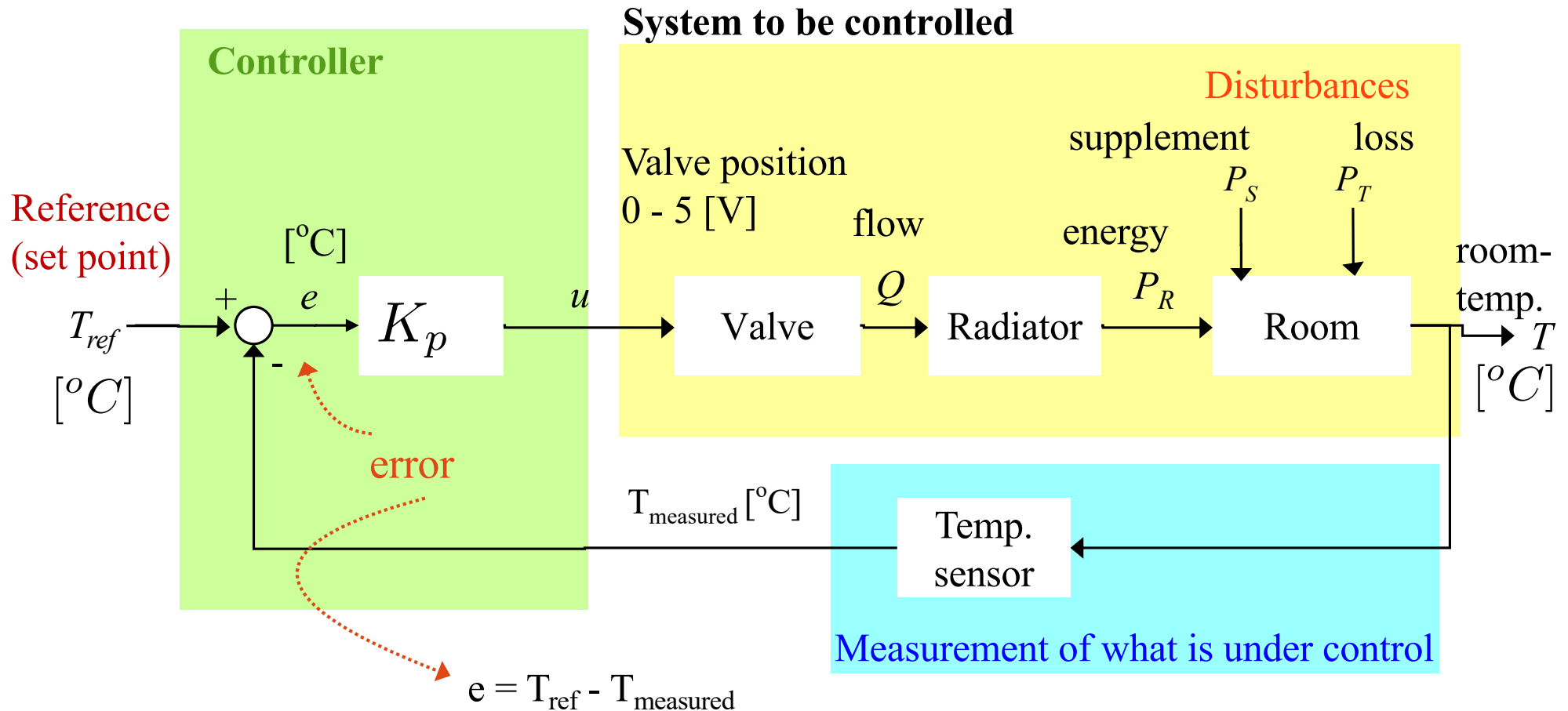
A P-controller uses **feedback** by adjusting the control signal based on the **error** between the reference and the measured output.

P-Controller

Closed-loop Controller

$$u = 0 \rightarrow T_{\text{measured}} = 10^{\circ}\text{C}$$

$$u = 5 \rightarrow T_{\text{measured}} = 30^{\circ}\text{C}$$



$$u = K_p \cdot e \quad K_p = ?$$

Basic rule: Start with a reasonable initial value of K_p and gradually adjust it until the performance requirements are met.

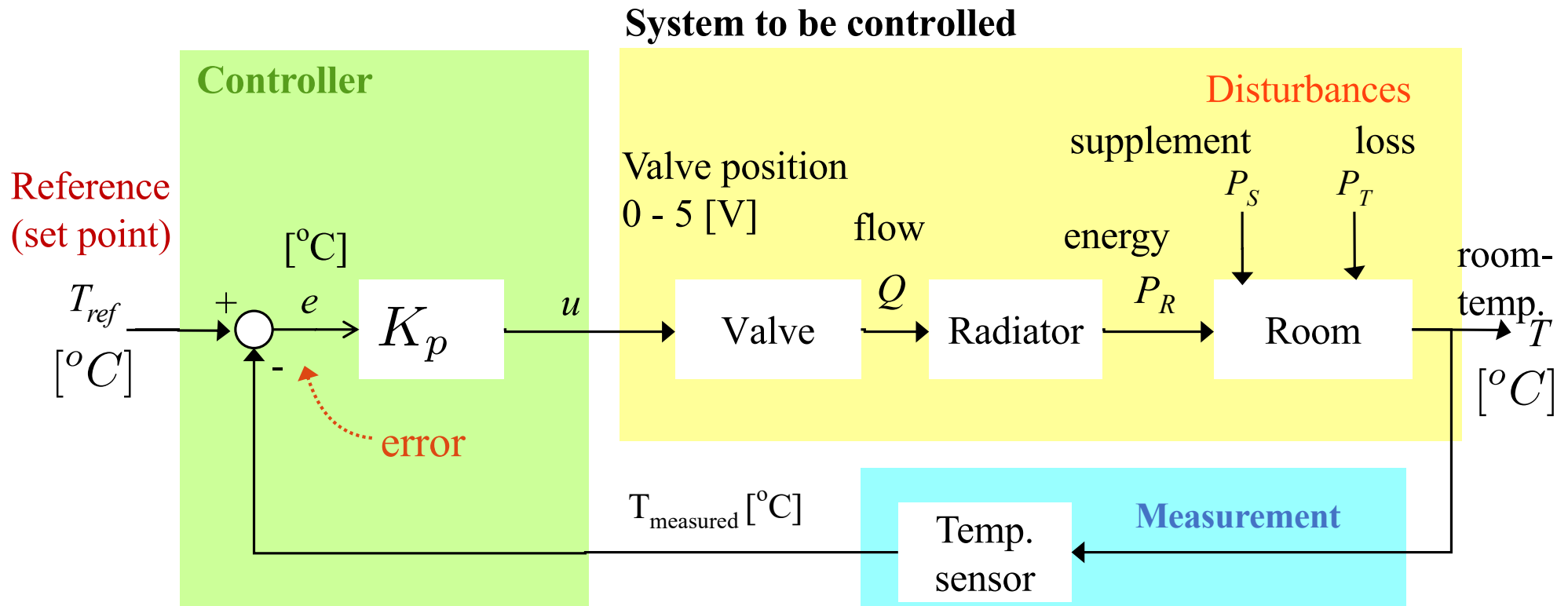
See Feedback control techniques book - Chapter 2. Hand-tuning

P-Controller

Closed-loop Controller

$$u = 0 \rightarrow T_{\text{measured}} = 10^{\circ}\text{C}$$

$$u = 5 \rightarrow T_{\text{measured}} = 30^{\circ}\text{C}$$



$$u = K_p \cdot e$$

$$e = T_{\text{ref}} - T_{\text{measured}}$$

$$K_p = ?$$

Example:

$$T_{\text{ref}} = 20^{\circ}\text{C}$$

$$\text{Demands: } u = 2.5 \text{ V}$$

$$\Rightarrow e = \frac{2.5}{K_p}$$

$\Rightarrow$

$$T_{\text{measured}} = T_{\text{ref}} - e$$

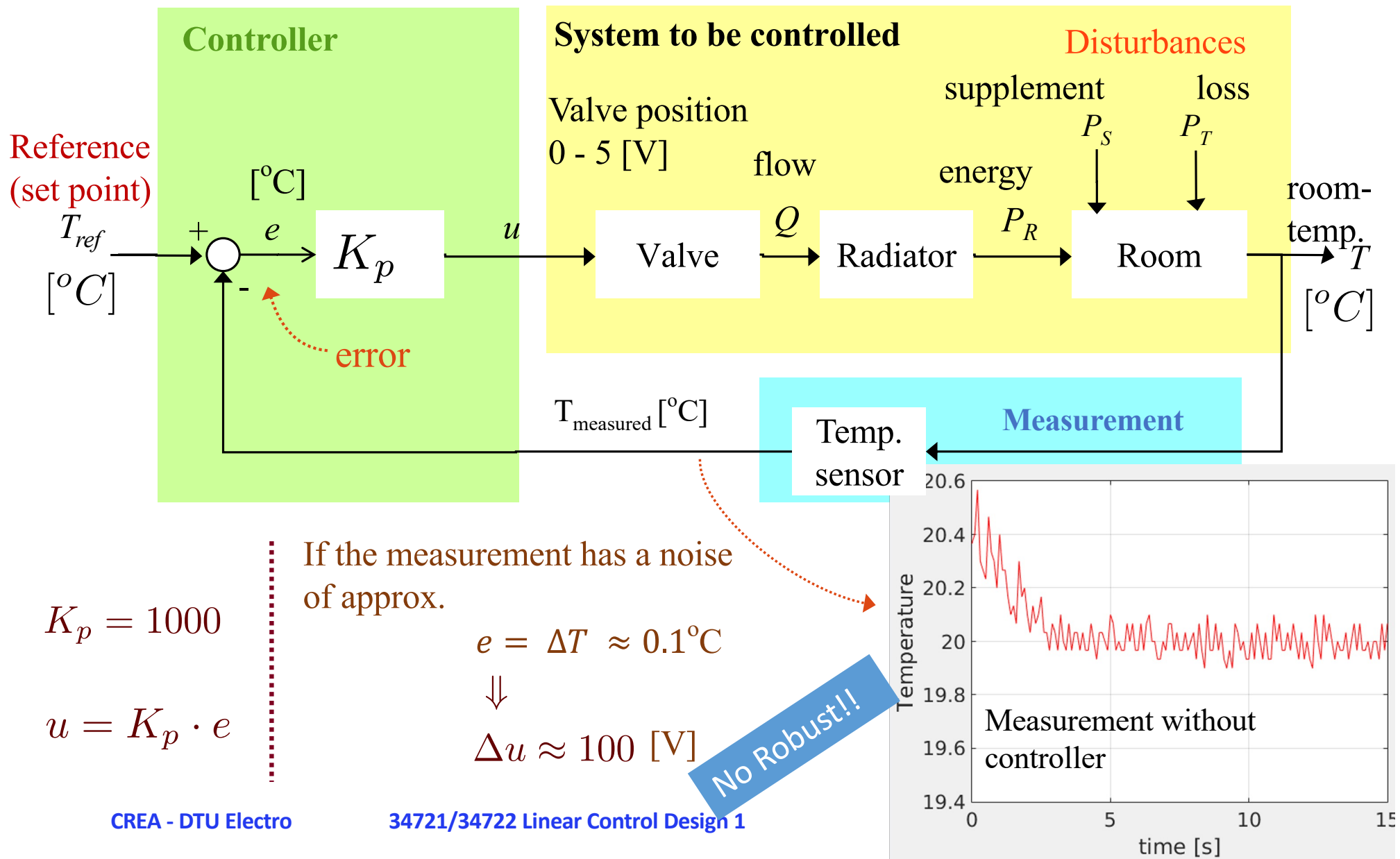
$$T_{\text{measured}} = T_{\text{ref}} - \frac{2.5}{K_p}$$

$$\lim_{K_p \rightarrow \infty} T_{\text{measured}} \rightarrow T_{\text{ref}}$$

Challenges?

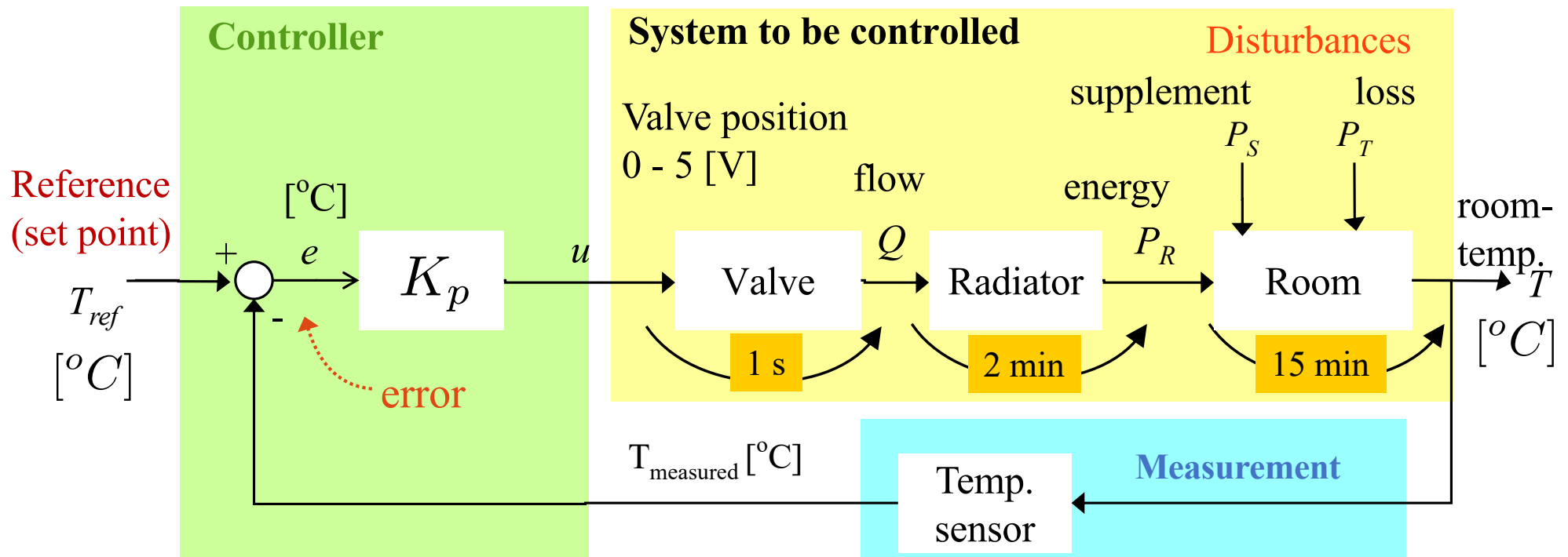
P-controller challenges: NOISE & LARGE K_p

Closed-loop Controller



P-regulator challenges: Stability & LARGE K_p

Closed-loop Controller



$$K_p = 1000$$

If it is a little too cold and K_p is great, u will open the **valve a lot** → it takes a short time (1s).

The flow heats the radiator slowly (2min)

The radiator emits an increasing power to the air near it.

After a while (15 min), the temperature has risen

e becomes smaller and the valve closes, however,

the radiator still emits a lot of energy, and the temp. continues to rise

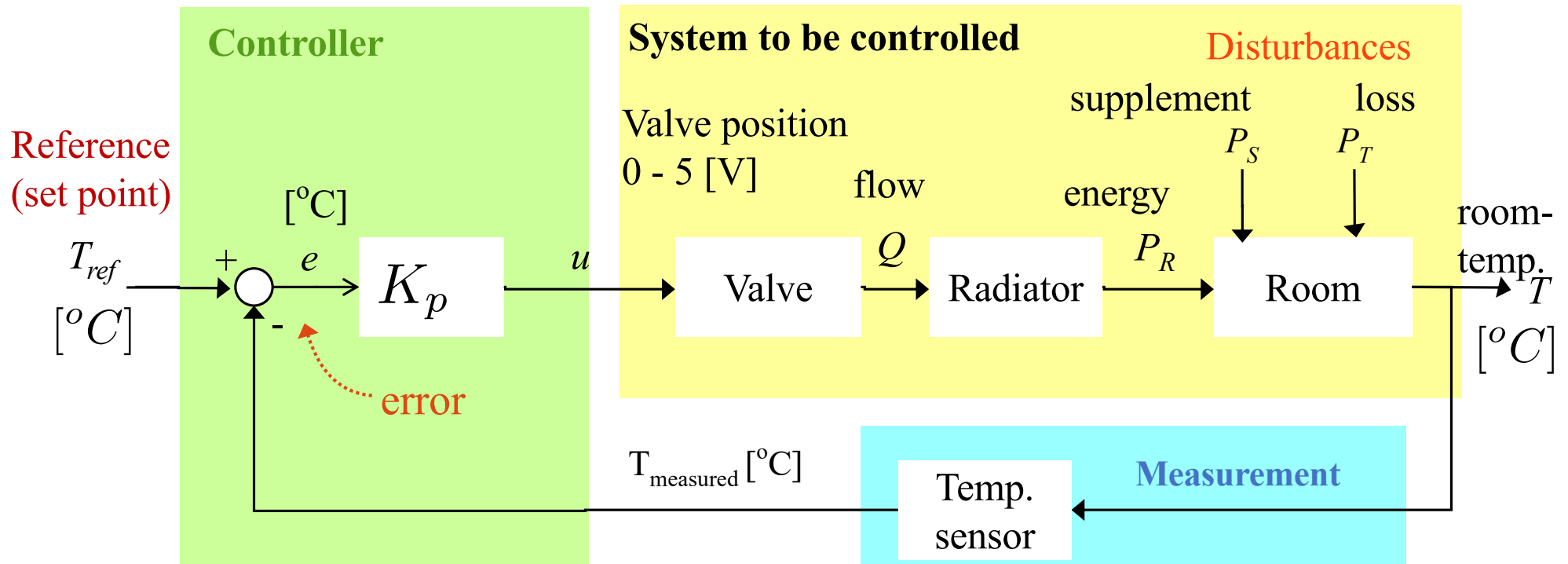
The valve closes for a long time, then repeats, **reinforcing instability!**

P-Controller – **SMALLER K_p**

Closed-loop Controller

$$u = 0 \rightarrow T_{\text{measured}} = 10^\circ\text{C}$$

$$u = 5 \rightarrow T_{\text{measured}} = 30^\circ\text{C}$$



$$K_p = 1$$

$$T_{\text{measured}} = T_{\text{ref}} - \frac{2.5}{K_p}$$

$$\Rightarrow T_{\text{measured}} = T_{\text{ref}} - 2.5$$

2.5 °C error!

P-Controller

The P-controller generates an actuator signal u that is proportional to the control error, meaning that changes in u directly influence the system output.

Advantage

- ✓ Can suppress disturbances once properly tuned

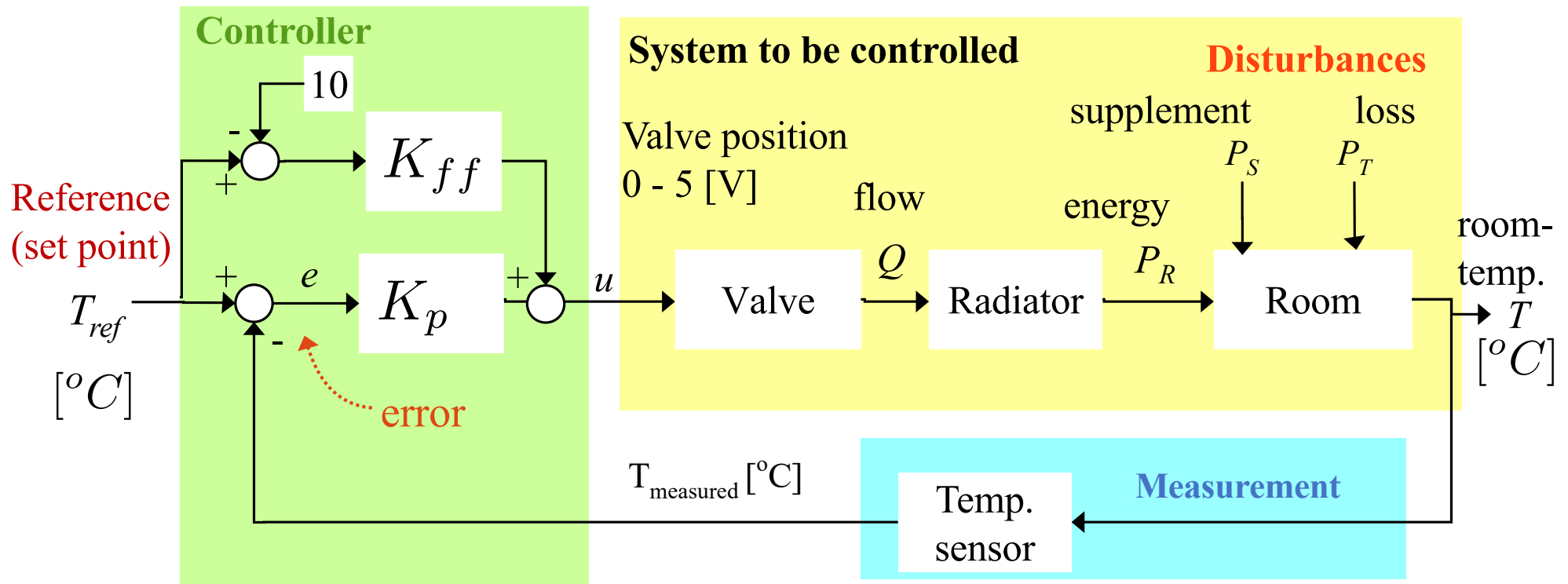
Disadvantage

- ✗ May result in a **steady-state (stationary) error** as $t \rightarrow \infty$, depending on the system type (more on this next lectures)
- ✗ Can cause **large control signals and oscillations**, especially in the presence of measurement noise
- ✗ High proportional gain can lead to **instability**

Combination Closed-loop Controller with feed forward

$$u = 0 \rightarrow T_{\text{measured}} = 10^{\circ}\text{C}$$

$$u = 5 \rightarrow T_{\text{measured}} = 30^{\circ}\text{C}$$



Problem

$$G_{ff} = 0.25(T_{\text{ref}} - 10)$$

$$K_p = 1$$

$$T_{\text{ref}} = 20$$

A **disturbance** causes 2 degrees lower than expected

$$T_{\text{measured}} = T_{\text{ref}} - 2 = (4u + 10) - 2$$

See next
slide

Problem

Only with the FF

$$u = G_{ff} = 0.25 \cdot (T_{ref} - 10)$$

$$u = 0.25 \cdot T_{ref} - 0.25 \cdot 10$$

$$T_{ref} = 4u + 10$$

A disturbance causes 2 degrees less:

$$T_m = T_{ref} - 2 = 4u + 10 - 2 \quad (a)$$

$$u = K_{ff} \cdot (T_{ref} - 10) = 0.25 \cdot (20 - 10) = 2.5$$

$$\text{Substitute } u \text{ in (a)} \rightarrow T_m = 4 \cdot (2.5) + 10 - 2 = \mathbf{18^\circ C}$$

FF + P-controller

$$u = K_{ff} \cdot (T_{ref} - 10) + K_p \cdot (T_{ref} - T_m)$$

$$u = 0.25 \cdot (20 - 10) + 1 \cdot (20 - T_m) = 2.5 + 20 - T_m = 22.5 - 4u - 10 + 2 = 14.5 - 4u$$

$$5u = 14.5 \rightarrow u = 14.5/5 = 2.9$$

$$\text{Substitute } u \text{ in (a)} \rightarrow T_m = \mathbf{19.6^\circ C \text{ (closer to } 20^\circ C!!!)}$$

Time-Domain Specifications

P-controller gain K_p

A higher gain can give a less **steady error**
(difference between desired and steady)

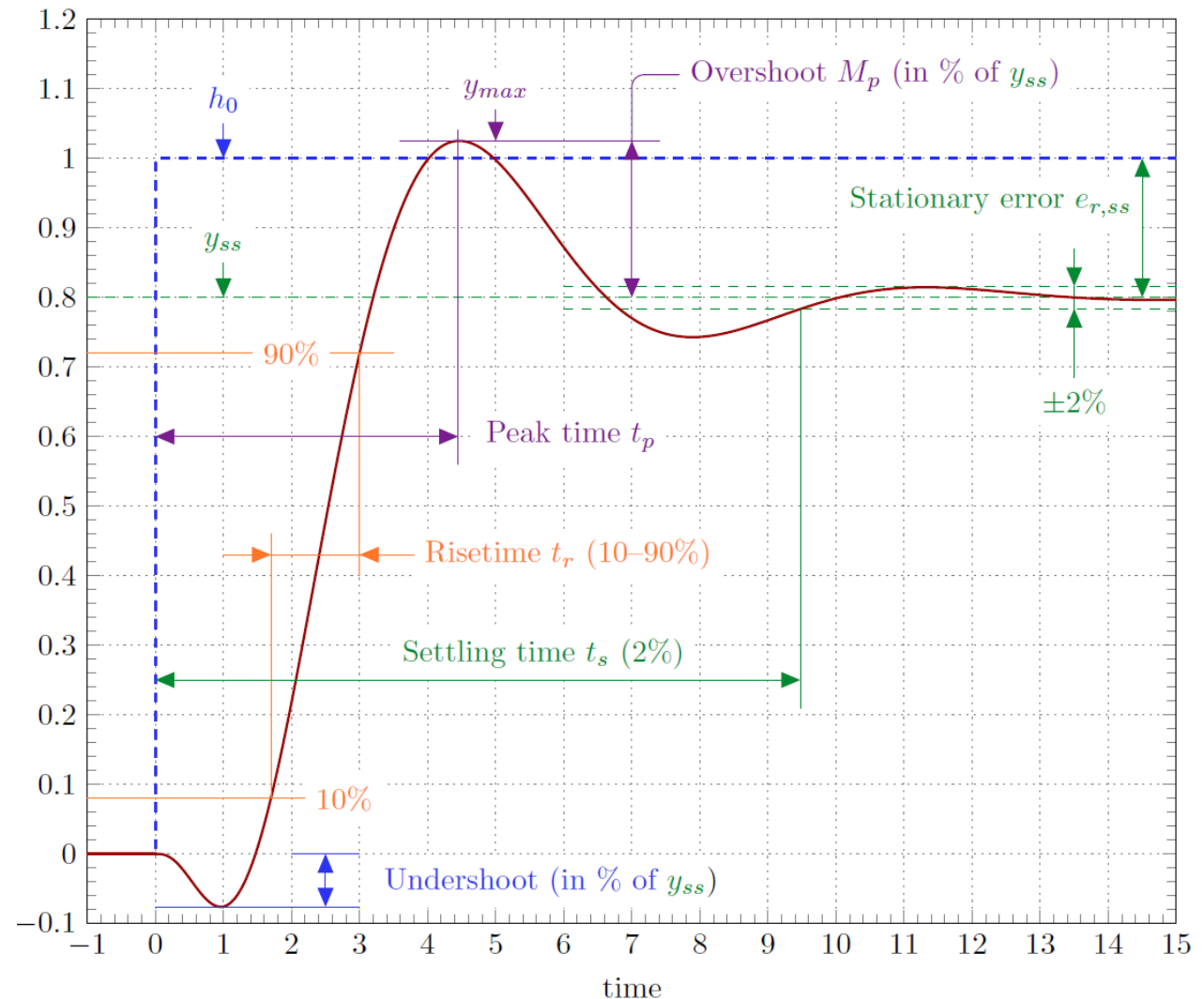
A higher gain can provide a faster **rise time**

A higher gain can cause **overshoot** and perhaps **instability**

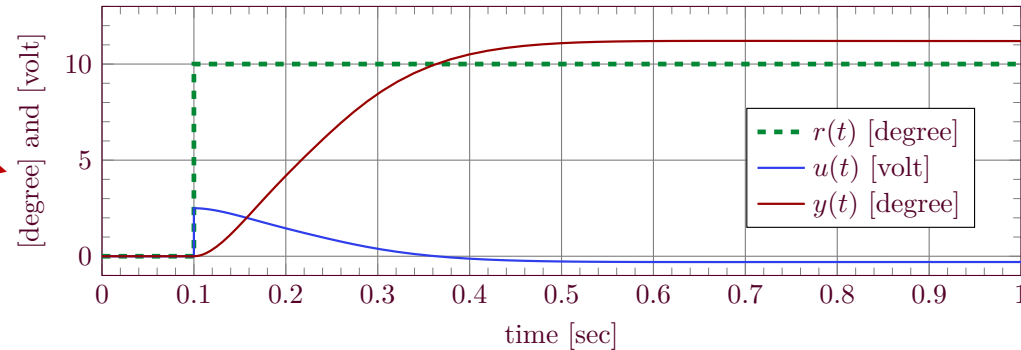
A lower gain can result in greater **steady error**

A lower gain results in a slower **settling time**

General specifications for signals

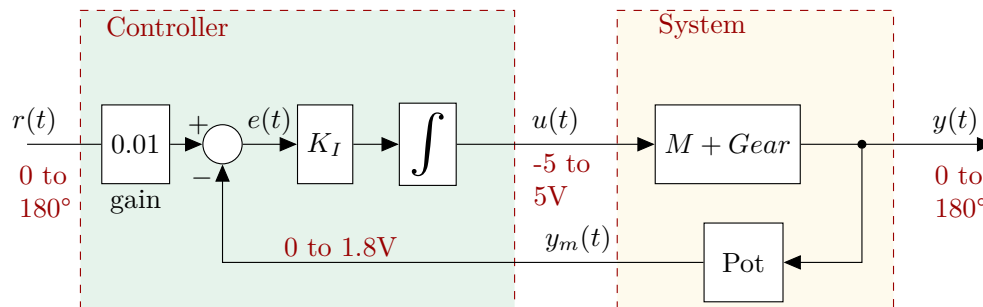


P vs I-Controller



P-controller for a DC servo system
Stationary error

$$u(t) = K_I \int e(t) dt$$

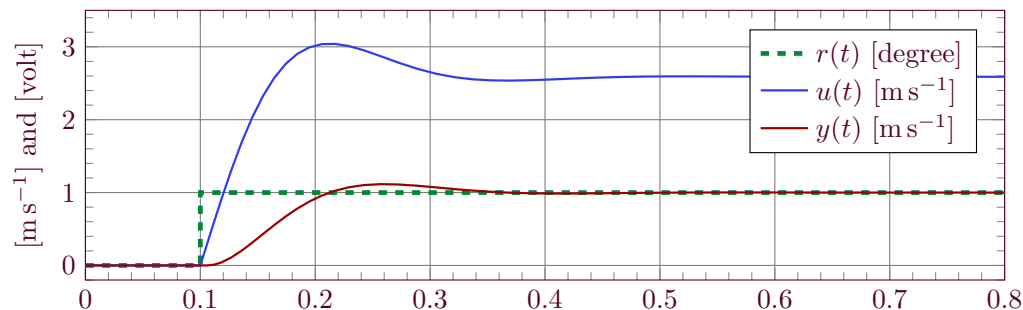


I-controller for velocity control

K_I determines how fast to integrate

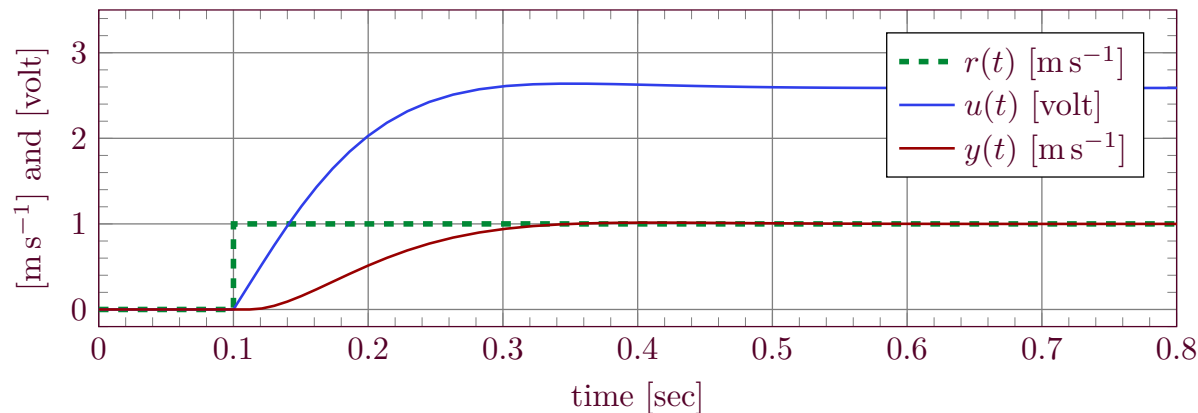
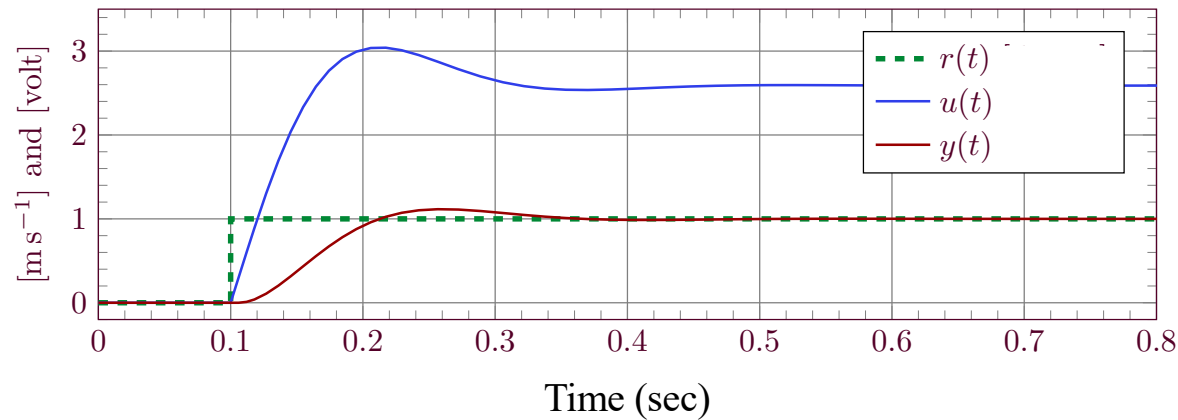
$$K_I = K_p \frac{1}{\tau_i}$$

τ_i reaction time, time to approach the desired value



I-Controller

$$u(t) = K_I \int e(t) dt$$



$K_i = 50$
 $\tau_I = 0.2$
overshoot

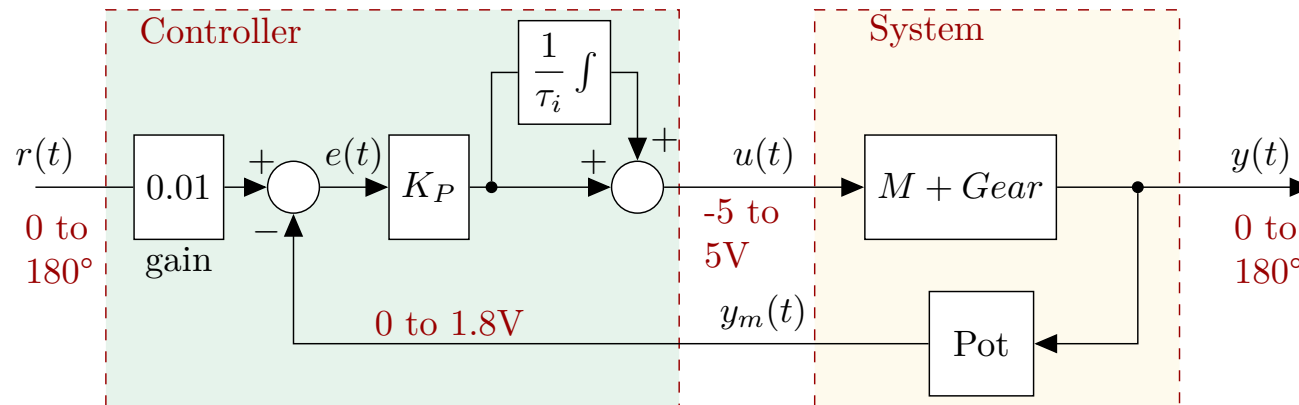
If there is an overshoot the constant can be made longer
➔ the response slower (less overshoot)

$K_i = 25$
 $\tau_I = 0.4$

➔ But the overshoot gets smaller

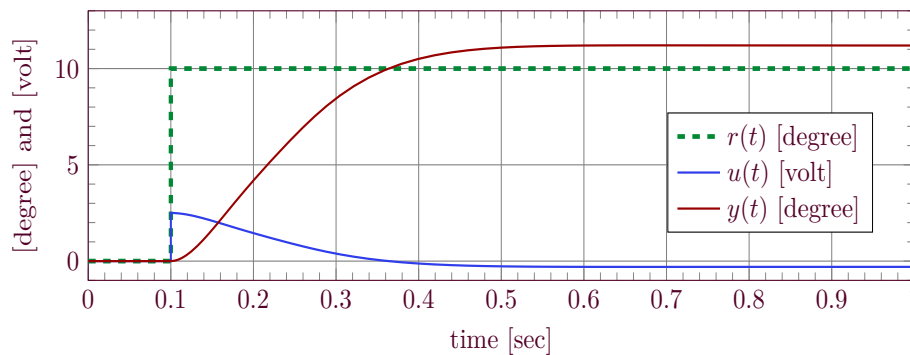
PI-Controller

$$u(t) = K_p(e(t) + \frac{1}{\tau_i} \int e(t)dt)$$

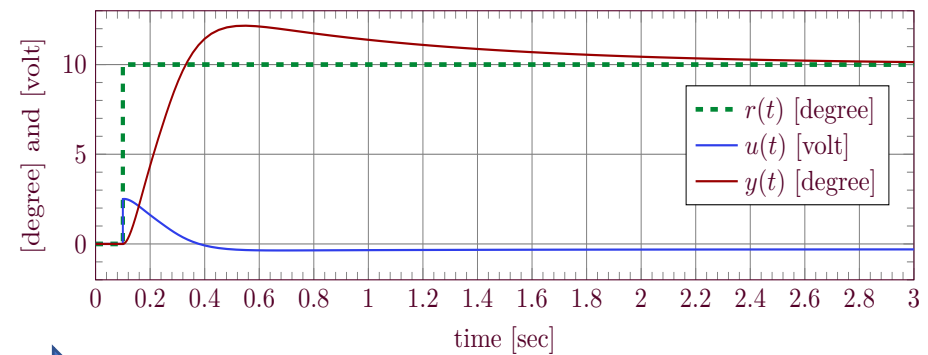


RECIPE

- First only P. The K_p value gives a faster response with no overshoot without the integration term.
- With I, we remove the stationary error but there is an overshoot given by the I. We refine it



Only P steady error

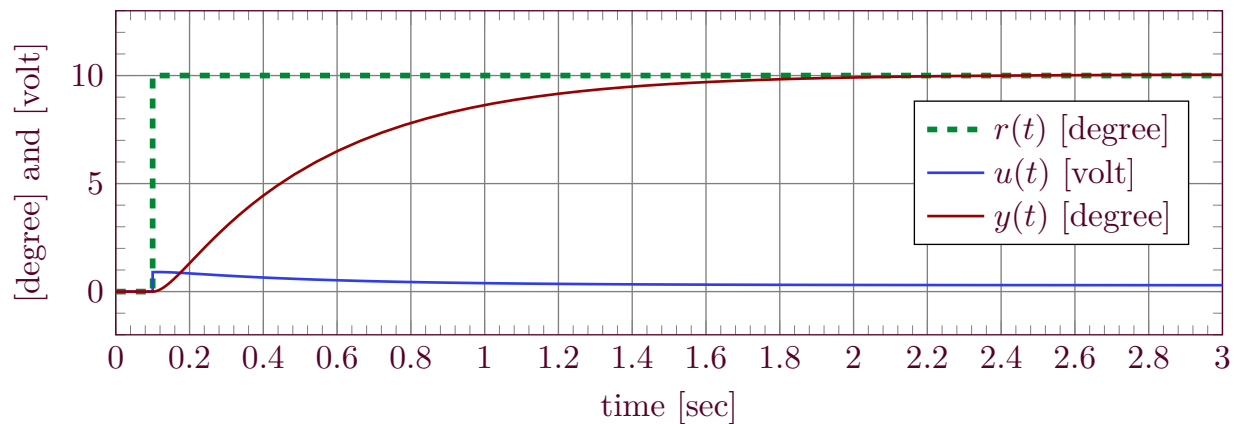


$$K_P = 25 \text{ V}/(^{\circ}/100)$$

$$\tau_i = 1 \text{ s}$$

PI-Controller Refinement

!! If the integration is too fast → possible instability
So, a slightly smaller K_p will usually allow for a faster τ_i



$$\begin{aligned} K_P &= 25 \text{ V}/(^{\circ}/100) \\ \tau_i &= 1 \text{ s} \end{aligned}$$



$$\begin{aligned} K_P &= 9, \\ \tau_i &= 1.4 \text{ s}, \end{aligned}$$

Reduce K_p and increase the τ_i

Summary: Hand-tuning PI

P-gain K_p $\rightarrow$ Initial $K_p = u_0/e_0$ $\leftarrow$ from a reasonable input change (step size or error e_0) to an estimate of a reasonable control signal u_0 .

τ_i could have a start value that is about the same time as it takes the output to reach a steady value.

The **refinement** can be based on the step-response:

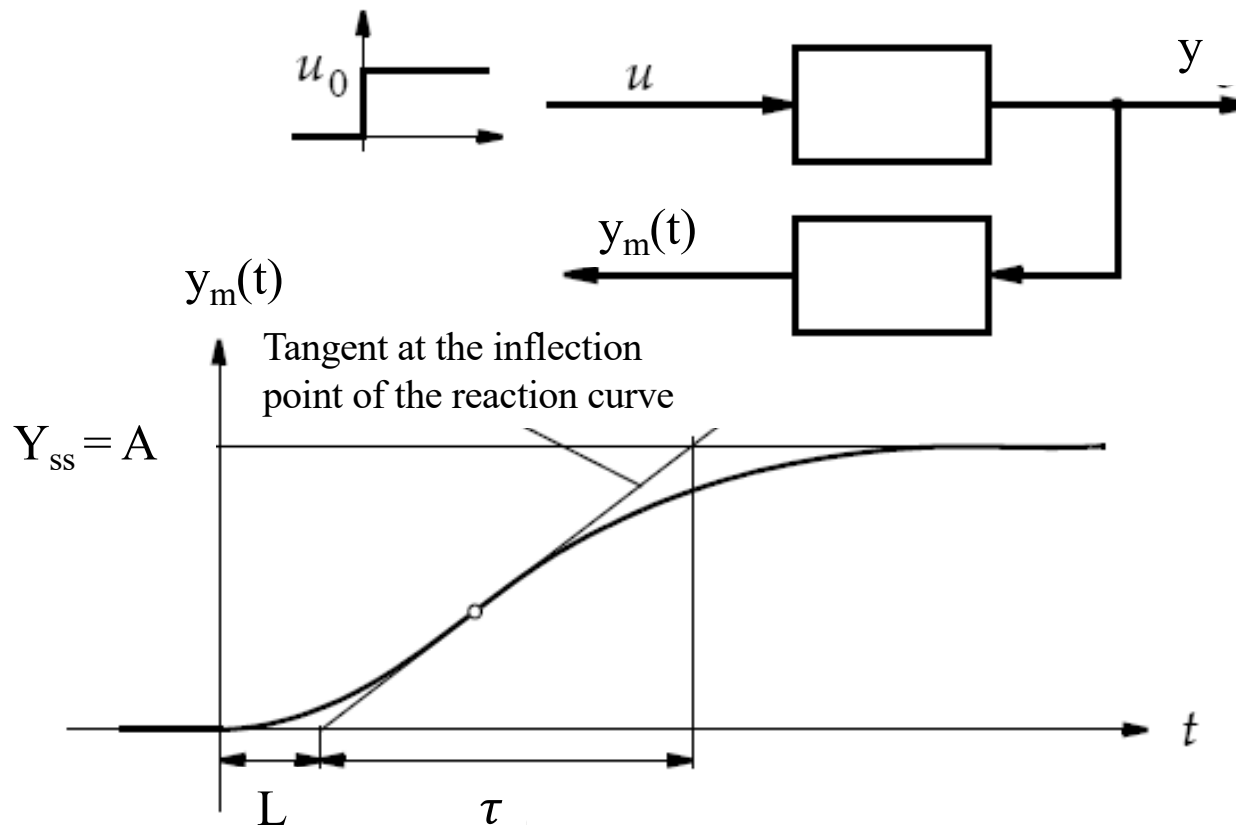
❖ **Decrease K_p** $\rightarrow$ more stable system with less overshoot or oscillations
Increase K_p $\rightarrow$ faster and more accurate system

❖ **Decrease τ_i** $\rightarrow$ to remove the stationary error faster
Increase τ_i $\rightarrow$ to get a more stable system

Ziegler-Nichols Tuning of the PID Controller

OPEN-LOOP

Chapter 3 Textbook



$$\text{Slope } R = \frac{A}{\tau} = \frac{1}{\tau} \frac{\Delta Y}{\Delta u}$$

Reaction time

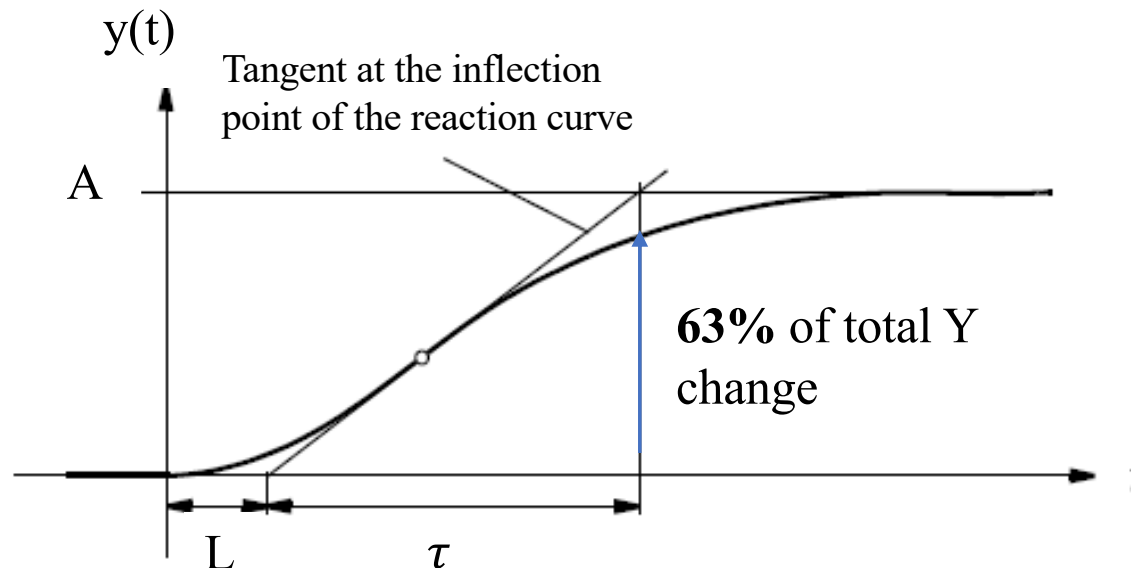
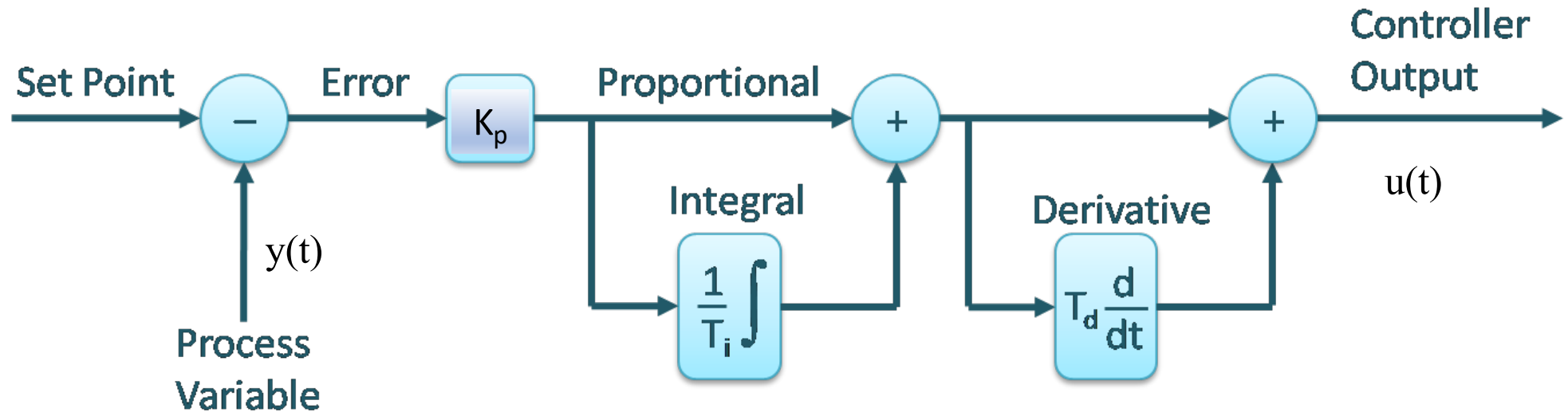
$$L = t_d \quad \text{Time delay or lag}$$

$$\tau \quad \text{Time constant (sec)}$$

$$\frac{Y(s)}{U(s)} = \frac{A e^{-sL}}{\tau s + 1}$$

A steady – state gain

OPEN-LOOP Tuning Procedure



- Make a step change in $u(t)$
- Calculate the Process Gain:

$$GP = \text{change in } Y / \text{change in } U$$
- Find the maximum slope on the response curve (inflection point) R
- Draw a tangent and determine time delay (L)
- Calculate the value of Y at 63% of the total change.
- Measure the time response (τ)

Ziegler-Nichols Tuning of the PID Controller

OPEN-LOOP

$L = t_d$ Time delay
 τ Time constant
 A Steady-State value

PID Type	K_p	T_i	T_d
P	$\frac{1}{RL}$		
PI	$0.9 \frac{1}{RL}$	$\frac{L}{0.3}$	
PID	$1.2 \frac{1}{RL}$	$2 L$	$0.5 L$

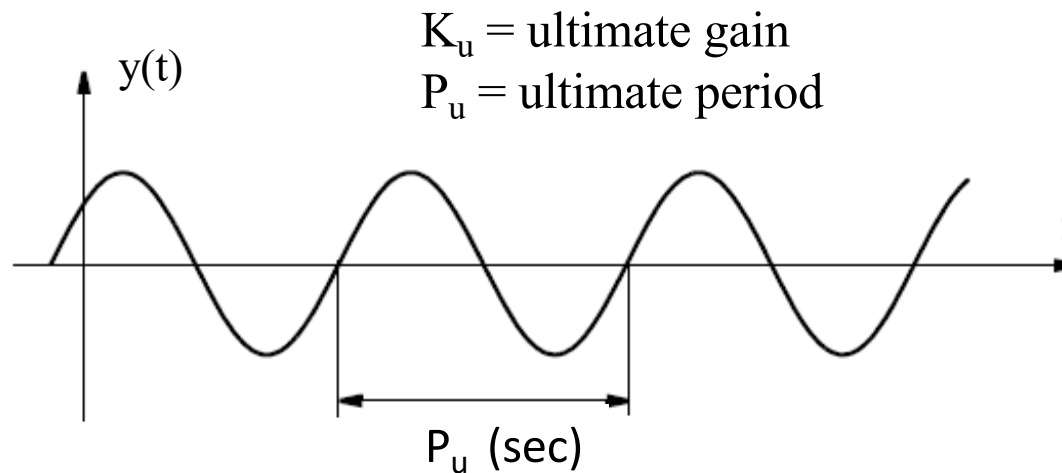
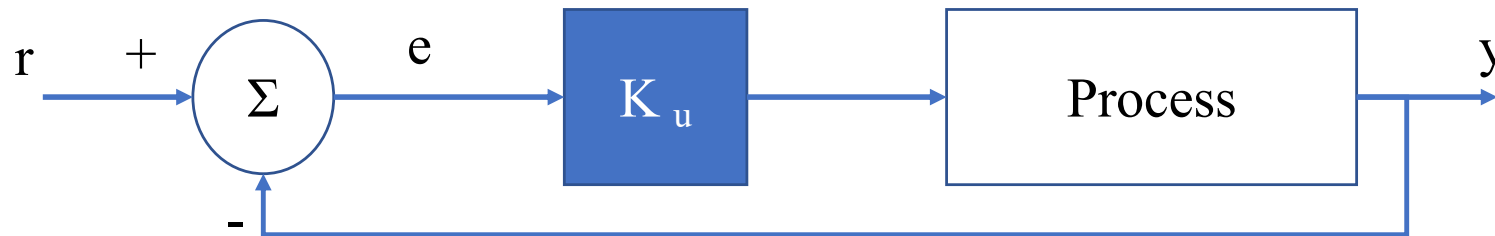
$$\text{Slope } R = \frac{A}{\tau} = \frac{1}{\tau} \frac{\Delta Y}{\Delta u}$$

$$\frac{Y(s)}{U(s)} = \frac{A e^{-sL}}{\tau s + 1}$$

Controller parameters according to Ziegler-Nichol's open-loop rules

Ziegler-Nichols Tuning of the PID Controller

CLOSED-LOOP



- Use STEP input for the reference
- Start controlling with a constant gain K
- Increase K to have a **continuous** oscillation response
- Choose the control gain and measure the period
- Design the control system (see table next slide)

Ziegler-Nichols Tuning of the PID Controller

CLOSED-LOOP

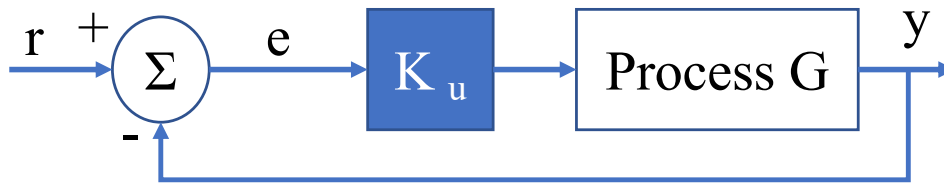
PID Type	K_p	T_i	T_d
P	$0.5 K_u$		
PI	$0.45 K_u$	$\frac{P_u}{1.2}$	
PID	$1.6 K_u$	$0.5 P_u$	$0.125 P_u$

Controller parameters according to Ziegler-Nichol's closed-loop rules

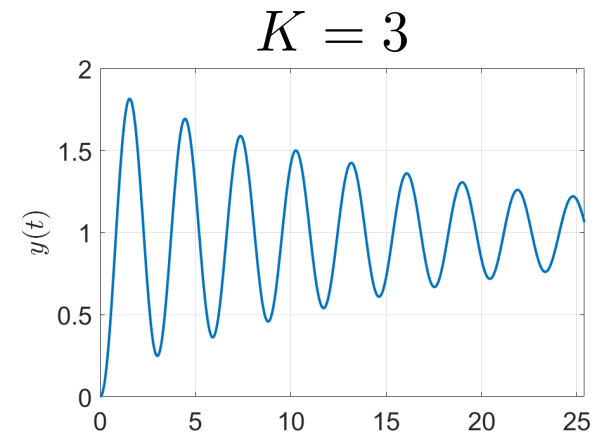
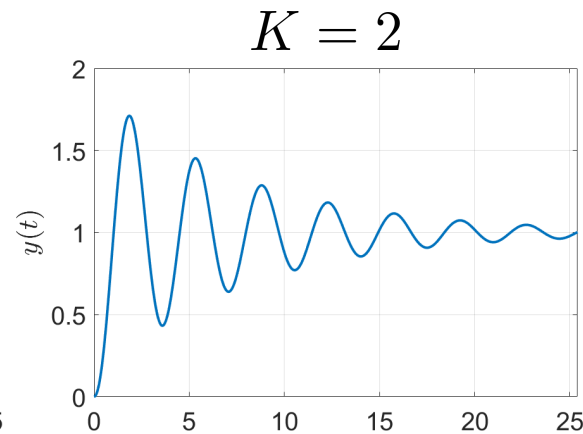
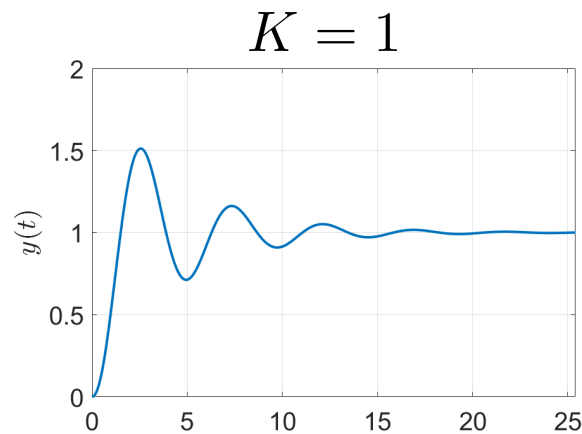
Ziegler-Nichols Tuning of the PID Controller

CLOSED-LOOP

Example $G(s) = \frac{s + 4.5}{s(s + 1)(s + 2)} = \frac{s + 4.5}{s^3 + 3s^2 + 2s}$



$$G_{cl}(s) = \frac{GK}{1 + GK}$$

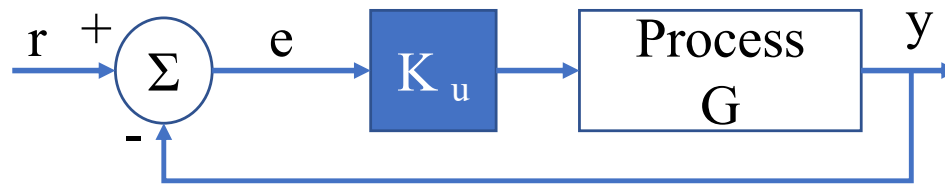


Ziegler-Nichols Tuning of the PID Controller

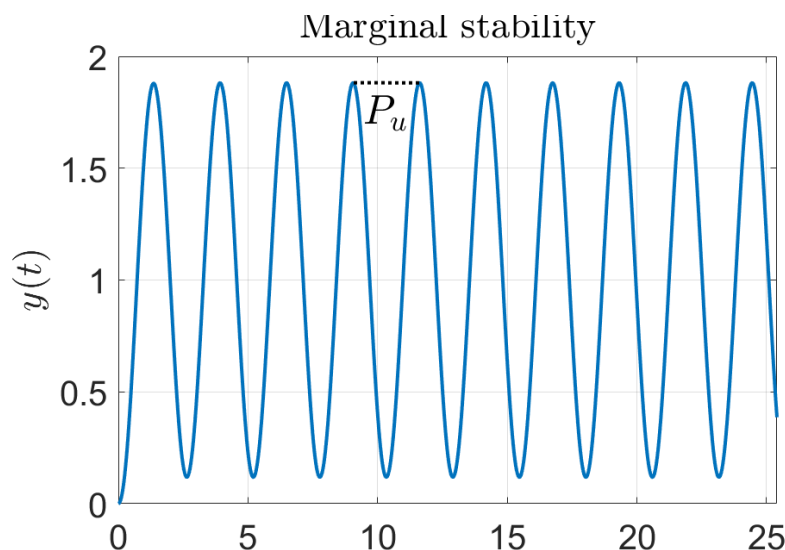
CLOSED-LOOP



Example $G(s) = \frac{s + 4.5}{s(s + 1)(s + 2)} = \frac{s + 4.5}{s^3 + 3s^2 + 2s}$



$K = 4$



$K = K_u = 4, P_u = 3 \text{ s}$

$k_p = 1.6K_u = 6.4$

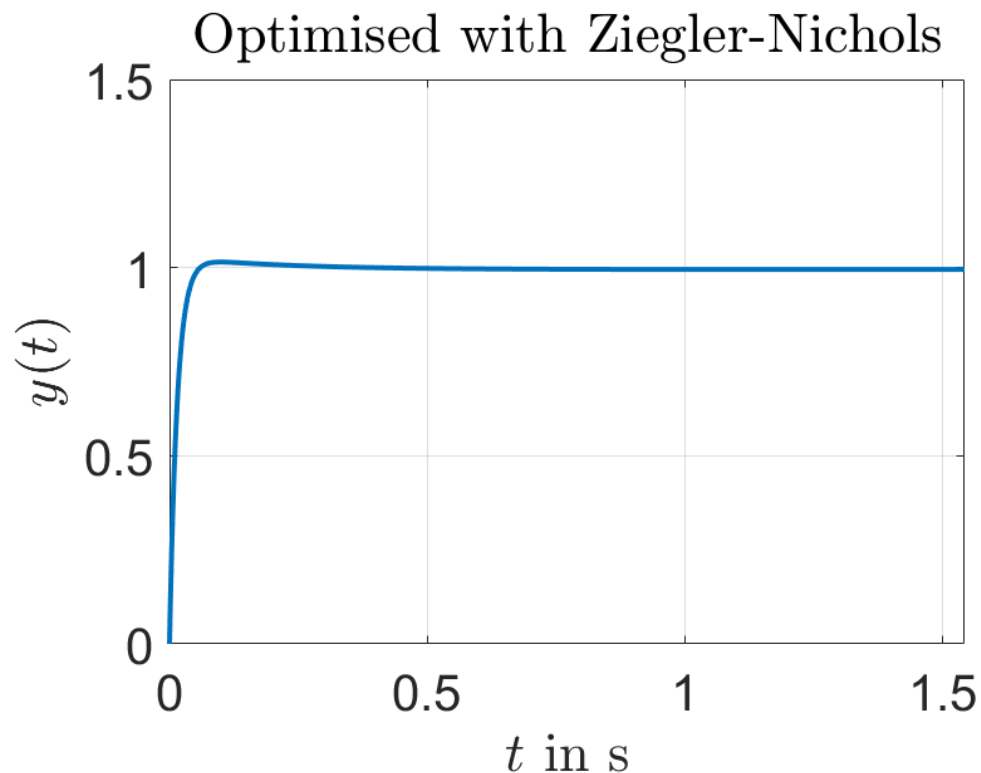
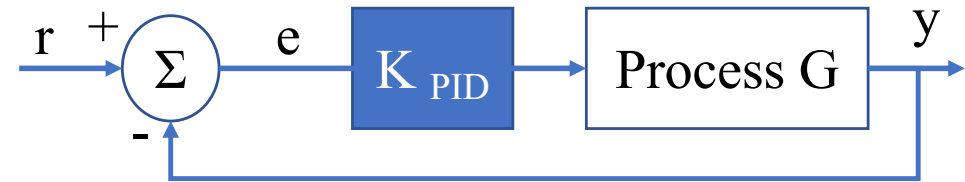
$T_i = 0.5P_u = 1.5 \text{ s}$

$T_d = 0.125P_u = 0.375 \text{ s}$

Ziegler-Nichols Tuning of the PID Controller

CLOSED-LOOP

Example



$$G_{cl}(s) = \frac{GK_{PID}}{1 + GK_{PID}}$$

$$K_{PID}(s) = k_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$k_p = 1.6K_u = 6.4$$

$$T_i = 0.5P_u = 1.5 \text{ s}$$

$$T_d = 0.125P_u = 0.375 \text{ s}$$

Ziegler-Nichols Tuning of the PID Controller

Chapter 3 - Textbook

- The methods are not optimal – **lack of generality**
- Open-loop methods can only be used with open-loop **stable** systems
- It might be problematic to make a closed-loop system **marginally stable**
- Difficult to calculate the needed/correct values in the open-loop case
- A good starting point for **further tuning**
- There exist other tuning rules but are dedicated to special types of systems